

DAVINCI

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Abstract. DAVINCI is the evolution of the Vocdoni voting protocol, designed to empower civil society by providing essential tools for secure, verifiable, and anonymous digital voting. Leveraging recent advancements in zero-knowledge (ZK) proof systems and blockchain technology, DAVINCI transitions to a specialized ZK-rollup system that inherits network security from settlement layers like Ethereum mainnet. The system relies on cryptographic proofs to ensure integrity and security, eliminating the need for centralized authorities. By integrating ZK-SNARKs and threshold homomorphic encryption, DAVINCI enables end-to-end verifiability, privacy, and trustlessness in the voting process. The protocol employs a distributed key generation among sequencers, coordinated via Ethereum smart contracts, and uses Ethereum data blobs for data availability. With a focus on accessibility, scalability, receipt-freeness, and automation, DAVINCI aims to facilitate high-frequency and low-cost voting, fostering mass adoption of e-voting and simplifying civil participation. Moreover, the introduction of the Vocdoni token aligns incentives among participants, ensuring the system's sustainability and enabling decentralized governance. Finally, the design of DAVINCI is grounded in practicality; all components are implemented using current technologies and have undergone proof-of-concept testing. In sum, the proposed architecture is not merely theoretical but a viable solution ready for short-term deployment.

- Consider separating the roles of the sequencers from key generators.
- Change (C_1, C_2) to **ciphertext** (in protocol flow and figures).
- Notation: are voteIDs called Nullifiers?

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1 Introduction

Online voting remains one of the most studied yet elusive applications in applied cryptography. As digital services expand across both public and private sectors, secure and universally verifiable online voting systems have become an increasingly desirable goal. Remote electronic voting promises scalability, accessibility, and administrative efficiency; yet the design of systems that simultaneously ensure privacy, integrity, coercion resistance, and transparency under realistic adversarial models remains a formidable challenge. Numerous deployments—including iVote, Helios, and the Estonian I-voting system—have revealed deep-rooted usability flaws and security gaps, particularly when verifiability mechanisms are either poorly understood by users or reliant on centralized infrastructure.

Against this backdrop, the Vocdoni project was initiated in 2018 with the aim of rethinking online voting from first principles. The name Voĉdoni, meaning “to give voice” in Esperanto, reflects the project’s foundational goal: to empower collectives—from small associations to millions of citizens—to engage in secure and verifiable decision-making, regardless of technological or institutional barriers. Central to this vision was the idea that voting is not limited to formal governmental elections but serves as a more general-purpose mechanism for collective signaling. Vocdoni introduced a fully anonymous, end-to-end verifiable voting system designed to operate efficiently across a range of devices, including smartphones. To support these goals, the team deployed a custom infrastructure emphasizing resilience, neutrality, and transparency.

Technically, the architecture of Vocdoni was grounded on a bespoke Byzantine fault tolerant (BFT) layer-1 blockchain, named Vochain. At the time, zero-knowledge (ZK) proof systems such as ZK-SNARKs were emerging but not yet practical for most deployments. Vochain provided a performant and low-cost environment (achieving approximately 700 transactions per second) where advanced cryptographic tools could be used without the constraints imposed by blockchains based on the Ethereum virtual machine (EVM). The ability to issue voting transactions without requiring user fees enabled broader accessibility. Over several years of development and deployment, this architecture proved both viable and valuable in practice. However, broader adoption as a universal voting protocol highlighted the need for further architectural refinements and stronger formal guarantees.

In this work, we introduce DAVINCI, a new protocol that builds upon the lessons and conceptual groundwork laid by Vocdoni. DAVINCI adopts a modular design and integrates state-of-the-art cryptographic tools, including succinct ZK proofs, improved bulletin board constructions, and robust coercion resistance mechanisms—reflecting the most recent advances in academic research. Unlike monolithic designs, DAVINCI is conceived as a composable primitive: a foundational layer intended to support secure and verifiable voting in diverse contexts, from blockchain governance to institutional elections. This shift in design philosophy aims to address long-standing challenges in the field, offering a cleaner abstraction with clearer security boundaries and formal underpinnings.

1.1 Contributions

This work introduces DAVINCI, a modular and verifiable protocol for digital voting, designed to support privacy-preserving, auditable, and adaptable election processes across diverse governance contexts. The protocol models voting as a constrained state machine, in which each valid operation is expressed as a formally defined state transition, enforced through succinct ZK proofs. That is, at the core of the system lies a set of reusable arithmetic circuits that implement voter authentication, encrypted ballot validation, state transitions, and tally finalization. These circuits are optimized for off-chain proving and are compatible with modern ZK-SNARK systems. Application-layer state is maintained off-chain and committed on-chain using cryptographic accumulators, specifically Merkle trees. The protocol defines two such trees: one for the voter registry (the census) and one for the voting state. These trees enable voters and verifiers to produce efficient inclusion and transition proofs, while minimizing on-chain storage and computation.

To coordinate protocol execution, DAVINCI includes a set of Ethereum smart contracts that mediate the setup of elections, manage finalization procedures, and verify submitted proofs of correctness. A standardized transaction and data encoding format supports all stages of the process, including vote casting, aggregation, and result publication. This approach ensures that protocol interactions remain deterministic and interoperable across clients and backends. The system also introduces the Vocdoni token, which is used

to align incentives between participants. Altogether, the design aims to balance auditability and privacy while reducing trust assumptions on infrastructure providers. By cleanly separating concerns and providing modular interfaces, the protocol is intended to serve as a foundation for both practical deployments and future extensions, such as integration with alternative identity systems, off-chain data availability layers, or other consensus backends.

- The Vocdoni voting protocol.
- The Vocdoni ballot protocol.

1.2 Related work

- State of the art of e-voting: <https://azkr.org/blog/evoting-review/>.

1.3 Paper organization

- Section 2: background.
- Section 3: high overview of the election process. We omit technical details.
- Section 4: cryptographic primitives used in the voting process. A description of the primitives but also the specific protocols being used and the details of their instantiation.
- Section 5: description of the voting process.
- Section 6: a proposed protocol for ballots. The rules of this protocol are implemented in the process management smart contracts.
- Section 7: the Vocdoni token.
- Section 8: analysis of the proposed protocols. It includes a security discussion of the properties of the protocol, implementation details, and a performance evaluation.
- Section 9: we finish with some last remarks.

2 Background

2.1 Interplanetary file system

- Brief introduction to IPFS.

2.2 Ethereum

- Ethereum blockchain (say transactions have a fee cost).
- Ethereum smart contracts.
Ethereum smart contracts are employed to orchestrate the voting process, manage state transitions, and store critical data, such as the current state root and encryption public keys. These smart contracts provide a trustless environment, ensuring that all participants can be confident that the voting process rules are correctly enforced.
- Ethereum data blobs.
EIP-4844 is used for data availability, enabling the storage of state transition data in the form of Ethereum data blobs. This mechanism allows sequencers to collaborate on the construction and verification of the voting process state, ensuring efficient and decentralized data availability.

2.3 Zero-knowledge proofs

- Introduce ZK-SNARKs.
- Introduce arithmetic circuits (arithmetic circuit satisfiability).

3 Protocol intuition

All the voting process is governed by the rules encoded in three smart contracts: the *process management*, the *sequencer registry*, and the *results verification* smart contract. All these three contracts are deployed on the Ethereum blockchain, and ensure that the execution of all steps is transparent and unalterable. The protocol consists of 5 phases, as illustrated in Figure 1, and described below.

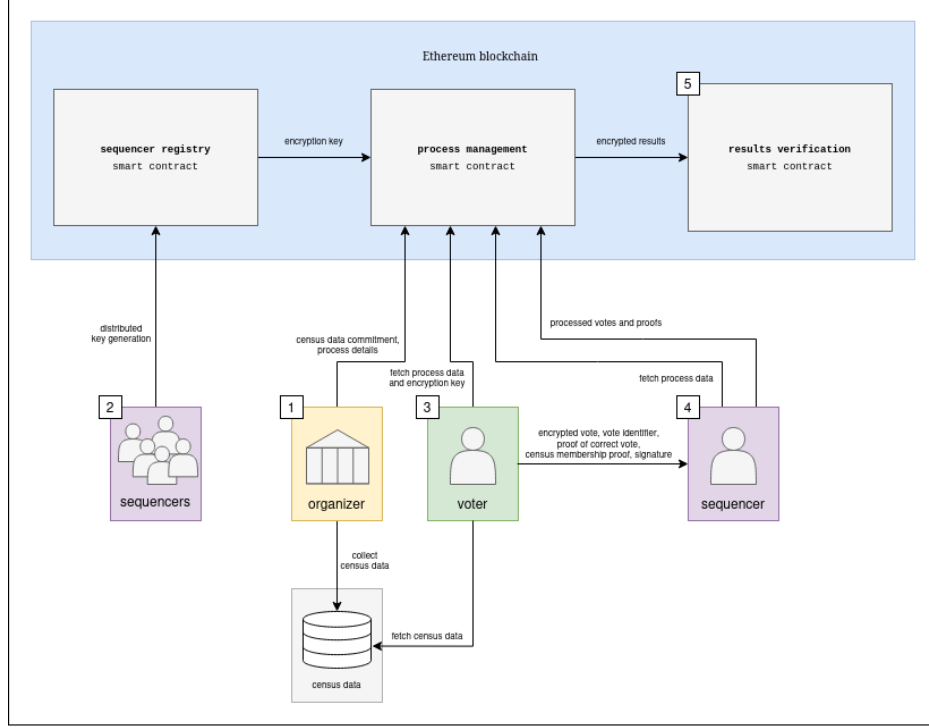


Fig. 1. Overview of the protocol flow.

1 Voting setup. In the first phase, the organizing entity gathers all census data and generates a cryptographic commitment to this data. Additionally, the organizer defines the voting parameters, including the number of voting options and the vote-counting mechanism (e.g., weighted voting, quadratic voting, etc.). Once these details are set, the organizer submits a transaction to the process management smart contract on the Ethereum blockchain. This transaction publicly records the voting parameters and census commitment, ensuring transparency and preventing any subsequent alterations to the voting setup.

2 Encryption key generation. A designated group of participants, referred to as *sequencers*, collaboratively generate a shared encryption public key, which voters will use to encrypt their votes. This key is established through a distributed key generation (DKG) protocol, ensuring that no single party can control or reconstruct the corresponding private key independently. Sequencers have a dual role in the protocol. In addition to generating the encryption key, they are also responsible for collecting votes from voters during the election. Consequently, their public key and the necessary information to contact them off-chain must be registered in the sequencer registry smart contract. Once the encryption public key is securely computed and published in the sequencer registry smart contract, the voting phase can start.

3 Voting. Voters select their preferred option according to the voting rules established by the organizer and captured in the process management smart contract. Instead of submitting their votes directly on-chain, they send them off-chain to a sequencer of their choice for processing. To ensure privacy, votes are encrypted

using the available encryption public key. Additionally, users compute a unique vote identifier that will allow them to verify that their vote has been included in the final result. Alongside the encrypted vote and the vote identifier, each voter must also provide the following data: a proof of valid voting, demonstrating that the vote complies with the election rules; a proof of eligibility, verifying that they are registered in the census; and a proof of identity ownership, in the form of a digital signature, confirming that they are the legitimate voter and are not impersonating someone else. Finally, to mitigate coercion and vote-buying, the protocol allows voters to overwrite their vote any number of times during the voting phase.

[4] Votes collection. During this phase, sequencers collect encrypted votes from multiple voters along with their corresponding proofs, and verify the validity of these submissions. That is, sequencers verify the signatures, to ensure the votes were cast by legitimate voters, they verify the proof of compliance, confirming that each vote adheres to the polling rules, and the census membership proofs against the commitment to the census data that was originally registered in the process management smart contract. Once the sequencers have processed and verified all votes, they must prove that these verifications were performed correctly. Instead of submitting individual verifications for each vote, they generate a single ZK proof that attests to the correctness of all verifications. Additionally, the sequencers reencrypt the votes and generate a proof of the correct reencryption computation. While this process does not alter the final tally, it prevents voters from decrypting their original vote. This step mitigates vote selling or coercion, as voters are no longer able to prove their choice to third parties. Finally, the sequencers submit the reencrypted votes along with their verification and reencryption proofs to the process management smart contract. The smart contract verifies all the proofs provided by the sequencers to check that they complied with the protocol.

[5] Results verification. At the conclusion of the voting period, the smart contract ceases to accept new state updates, effectively finalizing the process. The final state is then available on-chain for verification, providing an immutable record of the voting outcome.

4 Cryptographic primitives

In this section, we detail the cryptographic primitives used in Section 5. We briefly introduce elliptic curves, hash functions, Merkle trees, encryption schemes, and proof systems used in the protocol. We also specify at each step the concrete parameters with which each of the primitives is instantiated.

4.1 Elliptic curves

DAVINCI relies on multiple elliptic curves to ensure interoperability with Ethereum, compatibility with available cryptographic primitives, and efficient in-circuit operations. On the one hand, SECP256K1 [7] is used because Ethereum public keys are elements of this curve. As DAVINCI assumes each voter holds a standard Ethereum address, cryptographic operations such as digital signatures and identity verification rely on SECP256K1 to match the Ethereum ecosystem. On the other hand, BN254 [12] is chosen because it is the curve supported by Ethereum’s precompiled contracts for ZK proof verification, which makes it ideal for verifying SNARK proofs on-chain with minimal gas cost. Then, BabyJubjub [3] is used as an inner curve for elliptic curve operations within arithmetic circuits. Finally, BLS12-377 [6] and BW6-761 [8] are used to enable recursive proof composition. More specifically, BLS12-377 acts as the inner curve for constructing proofs, while BW6-761 serves as the outer curve that verifies those proofs within larger ZK-SNARK circuits. This pairing enables succinct verification and composability of ZK-SNARKs within other ZK-SNARKs, which we use for DAVINCI’s aggregation and state transition logic. Below, we give the details of these curves.

Parameters.

- $p = 0xfffefffffc2f$ (256-bit prime).
- $q = 0xfffffffffffffffffffffffffffffffebaaedce6af48a03bbfd25e8cd0364141$ (256-bit prime).
- $r = 0x2523648240000001ba344d8000000008612100000000013a700000000000013$ (254-bit prime).
- $s = 0x2523648240000001ba344d8000000007ff9f800000000010a1000000000000d$ (254-bit prime).
- $t = 0x60c89ce5c263405370a08b6d0302b0bab3eedb83920ee0a677297dc392126f1$ (251-bit prime).

- $u = 0x122e824fb83ce0ad187c94004faff3eb926186a81d14688528275ef8087be41707ba638e584e91903cebaff25b423048689c8ed12f9fd9071dcd3dc73ebff2e98a116c25667a8f8160cf8aeeaf0a437e6913e6870000082f49d00000000008b$ (761-bit prime).
- $v = 0x01ae3a4617c510eac63b05c06ca1493b1a22d9f300f5138f1ef3622fba094800170b5d44300000008508c00000000001$ (377-bit prime).
- $w = 0x12ab655e9a2ca55660b44d1e5c37b00159aa76fed00000010a11800000000001$ (253-bit prime).

Elliptic curve groups.

- SECP256K1 curve: $E^{\text{SEC}}/\mathbb{F}_p$ defined by equation $E^{\text{SEC}} : y^2 = x^3 + 7$, with group $\mathbb{G}^{\text{SEC}}$ of prime order q .
- BN254 curve: $E^{\text{BN}}/\mathbb{F}_r$ defined by equation $E^{\text{BN}} : y^2 = x^3 + 3$, with subgroups $\mathbb{G}_1^{\text{BN}}, \mathbb{G}_2^{\text{BN}}$ of prime order s , and an efficiently computable pairing $e : \mathbb{G}_1^{\text{BN}} \times \mathbb{G}_2^{\text{BN}} \rightarrow \mathbb{G}_T^{\text{BN}}$, where $\mathbb{G}_T^{\text{BN}} \subset \mathbb{F}_{s^{12}}$ and has order s .
- BabyJubjub curve: $E^{\text{BJ}}/\mathbb{F}_s$ defined by equation $E^{\text{BJ}} : 168700x^2 + y^2 = 1 + 168696x^2y^2$, with subgroup $\mathbb{G}^{\text{BJ}}$ of prime order t .
- BW6-761 curve: $E^{\text{BW}}/\mathbb{F}_u$ defined by equation $E^{\text{BW}} : y^2 = x^3 - 1$, with subgroups $\mathbb{G}_1^{\text{BW}}, \mathbb{G}_2^{\text{BW}}$ of prime order v , and an efficiently computable pairing $e : \mathbb{G}_1^{\text{BW}} \times \mathbb{G}_2^{\text{BW}} \rightarrow \mathbb{G}_T^{\text{BW}}$, where $\mathbb{G}_T^{\text{BW}} \subset \mathbb{F}_{v^6}$ and has order v as well.
- BLS12-377 curve: $E^{\text{BLS}}/\mathbb{F}_v$ defined by equation $E^{\text{BLS}} : y^2 = x^3 + 1$, with subgroups $\mathbb{G}_1^{\text{BLS}}, \mathbb{G}_2^{\text{BLS}}$ of prime order w , and an efficiently computable pairing $e : \mathbb{G}_1^{\text{BLS}} \times \mathbb{G}_2^{\text{BLS}} \rightarrow \mathbb{G}_T^{\text{BLS}}$, where $\mathbb{G}_T^{\text{BLS}} \subset \mathbb{F}_{w^{12}}$ and has order w .

Finite fields.

- $\mathbb{F}_p$: base field of the SECP256K1 curve.
- $\mathbb{F}_q$: scalar field of $\mathbb{G}^{\text{SEC}}$.
- $\mathbb{F}_r$: base field of the BN254 curve.
- $\mathbb{F}_s$: scalar field of $\mathbb{G}_1^{\text{BN}}, \mathbb{G}_2^{\text{BN}}, \mathbb{G}_T^{\text{BN}}$ and base field of the BabyJubjub curve.
- $\mathbb{F}_t$: scalar field of $\mathbb{G}^{\text{BJ}}$.
- $\mathbb{F}_u$: base field of the BW6-761 curve.
- $\mathbb{F}_v$: scalar field of $\mathbb{G}_1^{\text{BW}}, \mathbb{G}_2^{\text{BW}}, \mathbb{G}_T^{\text{BW}}$ and base field of the BLS12-377 curve.
- $\mathbb{F}_w$: scalar field of $\mathbb{G}_1^{\text{BLS}}, \mathbb{G}_2^{\text{BLS}}, \mathbb{G}_T^{\text{BLS}}$.

Generators.

We denote by G^{SEC} the generator of $\mathbb{G}^{\text{SEC}}$ as defined in [7], G^{BN} the generator of $\mathbb{G}_1^{\text{BN}}$ as defined in [17], G^{BJ} the generator of $\mathbb{G}^{\text{BJ}}$ as defined in [16], G^{BW} the generator of $\mathbb{G}_1^{\text{BW}}$ as defined in [11], and G^{BLS} the generator of $\mathbb{G}_1^{\text{BLS}}$ as defined in [15].

- $G^{\text{SEC}} = (0x79be667ef9dcbbac55a06295ce870b07029bfcd2dce28d959f2815b16f81798, 0x483ada7726a3c4655da4fbfc0e1108a8fd17b448a68554199c47d08fffb10d4b8)$.
- $G^{\text{BN}} = (0x01, 0x02)$.
- $G^{\text{BJ}} = (0x0b9fefffffffffaabff, 0x17fffffffffffffffffffebaedce6af48a03bbfd25e8cd0364141)$.
- $G^{\text{BW}} = (0x1075b020ea190c8b277ce98a477beaee6a0cfb7551b27f0ee05c54b85f56fc779017ffac15520ac11dbfcd294c2e746a17a54ce47729b905bd71fa0c9ea097103758f9a280ca27f6750dd0356133e82055928aca6af603f4088f3af66e5b43d, 0x58b84e0a6fc574e6fd637b45cc2a420f952589884c9ec61a7348d2a2e573a3265909f1af7e0dbac5b8fa1771b5b806cc685d31717a4c55be3fb90b6fc2cdd49f9df141b3053253b2b08119cad0fb93ad1cb2be0b20d2a1bafc8f2db4e95363)$.
- $G^{\text{BLS}} = (0x008848defe740a67c8fc6225bf87ff5485951e2caa9d41bb188282c8bd37cb5cd5481512ffcd394eeab9b16eb21be9ef, 0x01914a69c5102eff1f674f5d30afeec4bd7fb348ca3e52d96d182ad44fb82305c2fe3d3634a9591afd82de55559c8ea6)$.

4.2 Hash functions

DAVINCI uses different hash functions. On the one side, we use Keccak256 [4] for Ethereum address derivation, and Poseidon [9], MiMC and MiMC-7 [1] for arithmetic circuits within zero-knowledge proofs.

Keccak256. Keccak256 is the standard hash function used by Ethereum and is employed in DAVINCI to compress messages for signing and deriving Ethereum addresses from public keys over $\mathbb{G}^{\text{SEC}}$. Keccak256 takes arbitrary-length bitstrings and outputs 256-bit digests: $\text{Hash}_1 : \{0,1\}^* \rightarrow \{0,1\}^{256}$. This function is not SNARK-friendly and it is used exclusively in the authentication circuit, where the verification of Ethereum-compatible signatures requires reproducing the original message hash computed by Ethereum clients (see Section ??).

Poseidon. Poseidon is a SNARK-optimized hash function designed for efficient implementation inside arithmetic circuits. It is used in DAVINCI for computing commitments, nullifiers, and the state Merkle tree roots. Poseidon is used as $\text{Hash}_2 : \mathfrak{F}_s \rightarrow \mathbb{F}_s$, where $\mathfrak{F}_s$ is the set of tuples of $\mathbb{F}_s$ -elements of any length, and $\mathbb{F}_s$ is the field described in Section 4.1.

MiMC-7. Finally, we use functions from the MiMC family. On the one side, MiMC-7 hash function $\text{Hash}_3 : \mathbb{F}_p \rightarrow \mathbb{F}_p$, where $\mathbb{F}_p$ is the set of tuples of $\mathbb{F}_p$ -elements of any length, and $\mathbb{F}_p$ is the field described in Section 4.1. This hash is used to hash the public inputs of the different circuits and verify the associated proofs more efficiently (see Section 5.4 for further details). On the other side, MultiMiMC7. Different curves.

For the sake of clarity, in our notation we admit anything as an input of the above hash functions, although it might not be parsed as a sequence of field elements. Any bitstring can be converted to a tuple of field elements, so this does not change the actual working of the hash computation. In this document, it is implied that the proper conversion happens before feeding the input into the actual hash function.

4.3 Encryption schemes

To preserve ballot secrecy while enabling tallying, DAVINCI employs a threshold variant of the ElGamal cryptosystem instantiated over the BabyJubjub curve [14]. This scheme offers two properties crucial for the protocol: *additive homomorphism*, which allows the aggregation of encrypted votes without decryption, and *re-encryption*, which enables ciphertext randomization without altering the underlying plaintext. Together, these properties allow sequencers to tally votes while preventing voters from producing receipts that could be used for coercion or vote selling. The corresponding public key is generated collectively by the sequencers via the distributed key generation protocol described in Section 4.5, ensuring that no single entity can decrypt ballots unilaterally.

Encryption and decryption. Given a message m , the ElGamal encryption algorithm maps it into a group element M of $\mathbb{G}^{\text{BJ}}$ and outputs a ciphertext as follows:

- | | |
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| <ul style="list-style-type: none"> • $\text{Enc}_{\text{pk}}(m; k \in \mathbb{F}_t)$: 1. Map m to $\mathbb{G}^{\text{BJ}}$ via $M = m \cdot G^{\text{BJ}}$. 2. Compute $C_1 = k \cdot G^{\text{BJ}} \in \mathbb{G}^{\text{BJ}}$. 3. Compute $C_2 = k \cdot \text{pk} + M \in \mathbb{G}^{\text{BJ}}$. 4. Output $\text{ciphertext} = (C_1, C_2) \in (\mathbb{G}^{\text{BJ}})^2$. | <ul style="list-style-type: none"> • $\text{Dec}_{\text{sk}}(\text{ciphertext})$: 1. Parse $\text{ciphertext} = (C_1, C_2)$. 2. Compute $K = \text{sk} \cdot C_1 \in \mathbb{G}^{\text{BJ}}$. 3. Output $M = C_2 - K$. |
|--|--|

Since the plaintext message space is typically small, recovering m from M is efficient: although the mapping $m \mapsto M$ is not generally invertible, it can be reversed by brute-force search or optimized techniques such as baby-step giant-step [5].

Homomorphic addition and reencryption. ElGamal encryption is additively homomorphic. Given two ciphertexts (C_1, C_2) and (C'_1, C'_2) , their component-wise addition yields another valid ciphertext $(C_1 + C'_1, C_2 + C'_2)$. The resulting ciphertext decrypts to the sum of the two underlying messages. This property allows sequencers to aggregate encrypted ballots directly. Moreover, to prevent linkability and ensure receipt-freeness, sequencers also re-randomize ciphertexts. Re-encryption exploits this property by adding to a ciphertext an encryption of zero. This operation yields a fresh ciphertext of the same message under new randomness, making it computationally infeasible to link the re-encrypted ballot with the original submission.

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|--|--|
| <ul style="list-style-type: none"> • $\text{EncAdd}_{\text{pk}}(\text{ciphertext} \in (\mathbb{G}^{\text{BJ}})^2, \text{ciphertext}' \in (\mathbb{G}^{\text{BJ}})^2)$: <ol style="list-style-type: none"> 1. Parse $\text{ciphertext} = (C_1, C_2) \in (\mathbb{G}^{\text{BJ}})^2$. 2. Parse $\text{ciphertext}' = (C'_1, C'_2) \in (\mathbb{G}^{\text{BJ}})^2$. 3. Output $(C_1 + C'_1, C_2 + C'_2) \in (\mathbb{G}^{\text{BJ}})^2$. | <ul style="list-style-type: none"> • $\text{ReEnc}_{\text{pk}}(\text{ciphertext} \in (\mathbb{G}^{\text{BJ}})^2; k \in \mathbb{F}_t)$: <ol style="list-style-type: none"> 1. Parse $\text{ciphertext} = (C_1, C_2) \in (\mathbb{G}^{\text{BJ}})^2$. 2. Compute $\text{ciphertext}' = \text{Enc}_{\text{pk}}(0; k)$. 3. Output $\text{EncAdd}_{\text{pk}}(\text{ciphertext}, \text{ciphertext}')$. ■ |
|--|--|

In the protocol, re-encryption is applied not only to newly submitted ballots but also to ballots already stored in the state tree. By refreshing the randomness of both new and existing ciphertexts, sequencers ensure that it is indistinguishable whether a ballot has been overwritten or merely re-randomized. This mechanism is essential for guaranteeing receipt-freeness: voters cannot produce a verifiable receipt of their choice, and adversaries cannot detect or prove whether a particular vote has been replaced (see Section 8).

4.4 Digital signature schemes

DAVINCI uses the elliptic curve digital signature algorithm (ECDSA) [13] over the SECP256K1 curve to ensure compatibility with standard Ethereum wallets. Verification of ECDSA signatures is performed inside zero-knowledge proofs using a specialized circuit that emulates SECP256K1 arithmetic. This approach is necessary because SECP256K1 is defined over a 256-bit prime field that differs from the native field used in the ZK-SNARK circuit (see Section 5.4 for more details). Below, we describe the algorithms, which follow the standard ECDSA protocol. [\(The output of the hash does not match – check where variables live.\)](#)

- | | |
|---|--|
| <ul style="list-style-type: none"> • $\text{S.Sign}_{\text{sk}}(\text{message} \in \{0, 1\}^*)$: <ol style="list-style-type: none"> 1. Compute $h = \text{Hash}_1(\text{message}) \bmod q$. 2. Select a random scalar $k \xleftarrow{\\$} \mathbb{Z}_q^*$. 3. Compute $R = k \cdot G^{\text{SEC}} = (x_R, y_R) \in \mathbb{G}^{\text{SEC}}$. 4. Set $r = x_R \bmod q$. <ul style="list-style-type: none"> • If $r = 0$, go back to step 2. • Else, continue. 5. Compute $s = k^{-1} \cdot (h + r \cdot \text{sk}) \bmod q$. <ul style="list-style-type: none"> • If $s = 0$, go back to step 2. • Else, continue. 6. Output $\sigma = (r, s)$. | <ul style="list-style-type: none"> • $\text{S.Verify}_{\text{pk}}(\text{message} \in \{0, 1\}^*, \sigma = (r, s) \in (\mathbb{Z}_q^*)^2)$: <ol style="list-style-type: none"> 1. Check that $\text{pk} \in \mathbb{G}^{\text{SEC}}$. 2. Compute $h = \text{Hash}_1(\text{message}) \bmod q$. 3. Compute $w = s^{-1} \bmod q$. 4. Compute $u_1 = h \cdot w \bmod q$. 5. Compute $u_2 = r \cdot w \bmod q$. 6. Compute $X = u_1 \cdot G^{\text{SEC}} + u_2 \cdot \text{pk} = (x_X, y_X) \in \mathbb{G}^{\text{SEC}}.$ 7. Accept if $r = x_X \bmod q$; otherwise, reject. |
|---|--|

4.5 Key generation schemes

To ensure that no single entity controls the decryption key, DAVINCI employs a distributed key generation (DKG) protocol to jointly derive the ElGamal encryption key pair used for ballots encryption [?]. Sequencers who participate in the DKG ceremony are each identified by a unique index i . The outcome is a collective public key (**encryptionKey**) that is published on-chain and used by voters to encrypt their ballots, while the corresponding private key is secret-shared among the sequencers. Only a threshold number t out of n sequencers can later collaborate to decrypt the tally. Unlike classical DKG protocols where shares are exchanged in the clear, our construction uses encrypted shares and ZK-SNARK proofs of correctness. This allows the Ethereum smart contract to verify compliance without learning any of the underlying secrets, ensuring both security and verifiability in a fully decentralized setting.

Let t be the threshold parameter and n the number of sequencers, with $t \leq n$. Let G be a generator of the elliptic curve group of order q . Each participant P_i runs the following procedure.

- $\text{GenerateEncryptedShares}(i)$:
 1. Select random scalars $a_{i,0}, \dots, a_{i,t-1} \xleftarrow{\$} \mathbb{Z}_q$.
 2. Define the polynomial $f_i(x) = \sum_{j=0}^{t-1} a_{i,j} x^j$.
 3. For each $j \in \{0, \dots, t-1\}$, compute public commitments $C_{i,j} = a_{i,j} \cdot G$.
 4. For each sequencer P_ℓ with index $\ell \in \{1, \dots, n\}$, compute the secret share $s_{i,\ell} = f_i(\ell)$.

5. Encrypt each $s_{i,\ell}$ under P_ℓ 's public key using a simplified version of the EC integrated encryption scheme (ECIES), obtaining $EncECIES(s_{i,\ell})$.
6. Output $\{C_{i,j}\}_{j=0}^{t-1}$ and $\{EncECIES(s_{i,\ell})\}_{\ell=1}^n$, together with a ZK-SNARK proof attesting that the encrypted values are consistent with the commitments.

The smart contract verifies the encrypted shares.

- **VerifyEncryptedShare**($i, \ell, EncECIES(s_{i,\ell}), \{C_{i,j}\}$):
 1. Verify that the ZK-SNARK proof attached to $EncECIES(s_{i,\ell})$ is correct.
 2. If the proof is valid, the share is accepted; otherwise, the participant P_i may be slashed.

After collecting valid encrypted shares, each sequencer P_j can recover their secret share s_j by decrypting all contributions addressed to them:

- **DeriveSecretShare**(j):
 1. Decrypt all $EncECIES(s_{i,j})$ values received from other participants.
 2. Compute $s_j = \sum_{i=1}^n s_{i,j} \mod q$.
 3. Output s_j .

- **DerivePublicKey**($\{C_{i,0}\}_{i=1}^n$):
 1. Compute $pk = \sum_{i=1}^n C_{i,0}$.
 2. Output pk .

Note that $C_{i,0} = a_{i,0} \cdot G$, so the collective public key corresponds to $pk = s \cdot G$, where $s = \sum_{i=1}^n a_{i,0} \mod q$ is the collective private key, unknown to any single participant.

This approach ensures that the encryption public key is securely generated in a decentralized way, that each sequencer learns only their own secret share, and that the correctness of the entire DKG procedure is verifiable on-chain. Misbehavior, such as submitting invalid shares, can be detected and penalized through the slashing mechanism enforced by the smart contract (see Section 7).

4.6 Merkle trees

The DAVINCI protocol uses sparse Merkle Trees (SMTs) as its primary data structure for maintaining off-chain state. Following the construction in [2], these are full binary trees of fixed depth in which each leaf encodes a key-value pair and empty leaves are explicitly represented, allowing for both inclusion and non-inclusion proofs. A key property of SMTs is that the hash of each key determines a unique traversal path from the root to the leaf, regardless of insertion order. When inserting a new key, the tree descends until it encounters either an empty node or a leaf. In case of a path collision, the algorithm splits at the first differing bit, creating intermediate nodes accordingly.

In DAVINCI, SMTs are used to represent two core data structures: the census tree, which encodes the list of eligible voters along with their respective voting weights, and the state tree, which encodes the current status of submitted votes. Both trees have fixed depth 64. The census tree is instantiated with the MiMC hash function over the BLS12-377 scalar field, while the state tree uses Poseidon over the BN254 scalar field. Only the root hash of each tree is published on-chain, ensuring verifiability with minimal on-chain storage. To support state transitions and circuit verification, each tree MT has the following functions associated:

- **MT.Root()**: it returns the current Merkle root of the tree.
- **MT.Insert(path, leaf)**: it inserts a new leaf **leaf** at the position defined by the given path.
- **MT.Update(leaf_{old}, leaf)**: it updates an existing leaf in the tree.
- **MT.MembershipProof(leaf)**: it returns a Merkle proof of inclusion for the given leaf.
- **MT.NonMembershipProof(leaf)**: it returns a proof that the given leaf is not included in the tree.
- **MT.Verify(leaf, proof, root)**: it verifies that the given proof corresponds to the claimed root and leaf.

These functions are used by sequencers to construct proofs that certify the validity of state transitions. In particular, they allow for off-chain state evolution that is independently verifiable on-chain and within ZK circuits. Additional details on the structure of the census and state trees, as well as the encoding of their leaves, are provided in Section 5.2.

4.7 Zero-knowledge proof systems

Zero-knowledge succinct non-interactive arguments of knowledge (ZK-SNARKs) are a crucial component in ensuring the integrity of the voting process. Voters generate ZK-SNARK proofs to demonstrate that their encrypted ballots comply with the rules and constraints defined by the election parameters, without revealing any information about their choices. Similarly, sequencers produce proofs to certify the correctness of vote aggregation and state transitions throughout the protocol. [P.Prove \(\)](#), [P.Verify \(\)](#).

All ZK circuits in DAVINCI are compiled using the Groth16 proof system [10], a widely adopted ZK-SNARK construction known for its succinctness, efficient verification, and minimal proof size. Although all circuits rely on the same proving system, they are instantiated with different elliptic curves depending on the cryptographic requirements of each phase. Specifically, the vote validity circuit is compiled over the BN254 curve, the census-related circuit uses BLS12-377, and the aggregation circuit is instantiated over BW6-761. Finally, the last circuit used by the sequencer—responsible for generating the final proof of correct tallying and result—is compiled over BN254 as well, since this is the proof that is verified on-chain by the smart contract using Ethereum’s native precompiles. Further details on the circuits are provided in Section 5.4 and a summary of the instantiations can be found in Figure 3.

5 Voting protocol

This section outlines the DAVINCI voting protocol. We introduce the parties involved in the process in Section 5.1. In Section 5.2, we describe the census and state Merkle trees, and in Section 5.3 the Ethereum smart contracts. In Section 5.4, we present the ZK circuits used to enforce vote validity, authentication, aggregation, and state transitions. Finally, in Section 5.5, we detail the full protocol flow step by step, from the voting setup to the results validation and process finalization.

5.1 Parties involved

A voting process involves three types of participants: the organizer, the sequencer, and the voters.

Organizer. The organizer is the entity responsible for defining and setting up the voting process. Its key responsibilities include defining the voting parameters and gathering the census data. The organizer ensures that the election is structured correctly but does not participate in vote collection or tallying.

Sequencer. Sequencers are a set of decentralized parties that facilitate the voting process. On the one hand, they collaboratively generate a public encryption key that voters use to encrypt their votes. Then, during the voting phase, they receive and verify encrypted votes from voters and reencrypt votes to prevent coercion.

Voter. Voters are the participants that belong to the census that cast their votes in the election.

5.2 Merkle trees

Merkle trees are used to create a cryptographic representation of both the voter census registry and the voting process state.

5.2.1 Census tree

The census Merkle tree, denoted by $\text{MT}^{\text{census}}$, encodes the list of eligible voters and their associated voting weights. Formally, it is implemented as a sparse Merkle tree (SMT) of fixed depth¹, as described in Section 4.6.

¹In the present work, all circuits are instantiated using sparse Merkle trees as described above. However, the protocol does not strictly depend on this structure: any authenticated data structure that supports efficient commitments and membership proofs verifiable inside zero-knowledge circuits could be employed. Adapting the protocol to such alternatives would only require modifying the corresponding circuits, while the overall flow of the voting process would remain unchanged.

Each voter is mapped to a unique path in the tree, typically derived from their Ethereum address or another unique identifier, and the corresponding leaf stores the voting weight of that address. While the default weight is usually one, the system also supports weighted voting, where different participants may have different levels of influence.

To construct the census, the organizer collects the dataset of eligible voters and their weights, computes the corresponding tree, and publishes its root (**censusRoot**) on-chain. The full dataset does not need to be stored on-chain; instead, it is distributed off-chain (e.g., via IPFS or another content-addressed storage system), enabling verifiers and sequencers to independently reconstruct the tree and produce membership proofs when needed. This design ensures transparency and verifiability while minimizing on-chain storage requirements.

The census tree plays a central role during the voting phase. Voters must provide a Merkle proof showing that their address and weight are included in the tree committed by the organizer. These proofs are then verified within ZK circuits, which guarantees that only eligible participants can cast ballots without requiring the disclosure of the entire census. In this way, the census Merkle tree provides a compact, cryptographically verifiable commitment to voter eligibility that supports both privacy and scalability.

Although the protocol does not mandate a specific method for constructing the census tree, different instantiations are possible depending on the application context:

- *Private datasets.* Organizations maintaining internal voter registries (e.g., membership databases or CSV files) can directly convert them into the Merkle format. This approach keeps the registry confidential, with only the root hash made public.
- *Self-sovereign identity (SSI).* If voters manage identities through SSI systems, the organizer can aggregate these identities and build the tree accordingly. This enables participation without requiring centralized identity management.
- *Ethereum-based tokens.* Token balances (e.g., ERC-20 or NFTs) may also define eligibility. In this setting, the census can either be created via the optimistic approach, taking a snapshot of token balances and allowing a dispute period for fraud proofs, or through a ZK-SNARK that proves correctness directly from Ethereum storage proofs and ensures that both individual balances and the token’s total supply are consistent.

This flexibility allows the protocol to accommodate a wide range of use cases, from private associations to decentralized communities, while maintaining the same security and verifiability guarantees.

5.2.2 State tree

The state Merkle tree, denoted by MT^{state} , represents the evolving state of the voting process. Like the census tree, it is implemented as a SMT of fixed depth, as described in Section 4.6. In contrast to the census tree, which remains static once the election begins, the state tree is dynamic: it is updated throughout the process as new votes are cast, validated, and incorporated by the sequencers.

The leaves of the state tree store the global process parameters, the encrypted ballots, and the set of vote identifiers, as illustrated in Figure 2. More specifically:

Reserved addresses for metadata. The first six addresses are reserved for the process parameters as follows:

- **0x0**: it stores a unique identifier for the voting process (**processID**).
- **0x1**: it stores the committed census root (**censusRoot**).
- **0x2**: it stores the ballot mode configuration, which is the set of rules for validating votes (**ballotMode**).
- **0x3**: it stores the encryption public key used to encrypt the votes (**encryptionKey**).
- **0x4**: it stores an accumulator of the encrypted votes that need to be added to the tally (**resultsAdd**).
- **0x5**: it stores an accumulator of the encrypted votes that need to be subtracted from the tally because they have been overwritten (**resultsSub**).

Encrypted ballots. Each voter’s encrypted ballot is stored in the state tree at a leaf whose path corresponds to their Ethereum address (or another unique identifier). The leaf value is the ciphertext corresponding to the ballot generated under the election’s public encryption key. If the voter later overwrites their ballot, the same path is updated with the new ciphertext.

Vote identifiers. In addition to storing ballots, the state tree also tracks the set of vote identifiers. Each vote identifier `voteID`, which is derived from the `processID`, the voter’s address, and fresh randomness, determines the path to a dedicated leaf. Initially empty, these leaves are filled when the corresponding `voteID` is used, thereby recording that it has been consumed. This enables efficient non-membership proofs: before accepting a new ballot, the sequencer checks that the corresponding `voteID` path is still empty, and upon inclusion, the leaf is marked as used to prevent reuse.

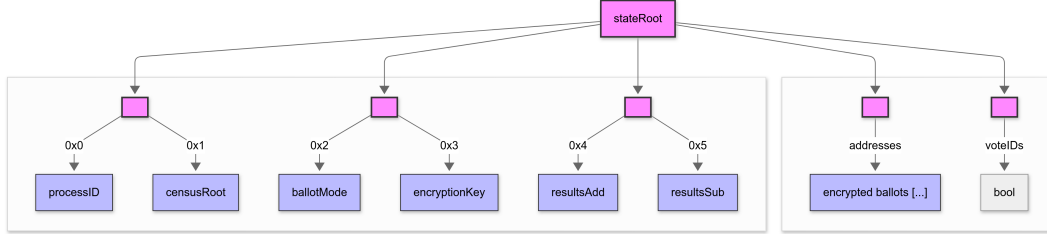


Fig. 2. Structure of the state Merkle tree. The first leaves are reserved for global process parameters while subsequent leaves store encrypted ballots indexed by voter addresses and vote identifiers.

At the beginning of the election, the state tree is initialized with the process parameters and empty accumulators, and its root (`stateRoot`) is published on-chain. As the voting progresses, sequencers update the tree whenever new votes are received, producing a new root for each transition. To guarantee correctness, each update is accompanied by a ZK-SNARK proof attesting that the transition from the previous root to the new root respects all protocol constraints: ballots are well-formed and comply with the voting rules, voters are eligible according to the census, the `voteID` non-membership condition prevents double voting (or overwrites are handled correctly), and encrypted results are aggregated consistently. In addition, sequencers reencrypt stored ciphertexts, including both newly submitted and already included ballots. This mechanism refreshes the randomness of the encrypted votes without altering their underlying content, ensuring receipt-freeness and concealing whether a particular ballot has been overwritten or simply re-randomized.

In this way, the state tree provides a compact and tamper-proof commitment to the current status of the election. At the end of the process, the final state root together with the on-chain verification of all transitions serves as a complete cryptographic record of the election, encapsulating both the tally and the integrity of the entire voting procedure.

5.3 Smart contracts

An Ethereum compatible network is used as the source of truth for the voting system. By leveraging an Ethereum virtual machine (EVM) blockchain, the voting process management ensures that all transitions are immutable and verifiable by all participants. To this end, we implement the following three smart contracts.

Process management smart contract. It is responsible for the life-cycle management of voting processes. It includes the initiation, monitoring, execution, and closure of voting events.

Sequencer registry smart contract. It keeps track of the existing available sequencers, stores the collateral of the sequencers to ensure (that incentivizes) good behaviour, and is used to coordinate the distributed key generation when a new voting process is created.

Results verification smart contract. For each voting process, this smart contract maintains the integrity of the cast votes and process life cycle. It verifies that each state transition committed by a sequencer adheres to the predefined rules.

5.4 Circuits

At the core of DAVINCI lie a set of arithmetic circuits that enforce the correctness of every step of the voting process. Each circuit corresponds to a distinct task and, taken together, they guarantee that all protocol rules are satisfied without revealing any private information. Specifically, the *ballot circuit* (Section 5.4.1) ensures that encrypted votes are valid and well-formed, the *verifier circuit* (Section 5.4.2) verifies voter eligibility and census membership; the *aggregation circuit* (Section 5.4.3) combines multiple authenticated votes into a single proof; and the *state transition circuit* (Section 5.4.4) updates the global state root and produces the ZK-SNARK proof that is verified on-chain. The flow of these circuits and their interactions is shown in Figure 3.

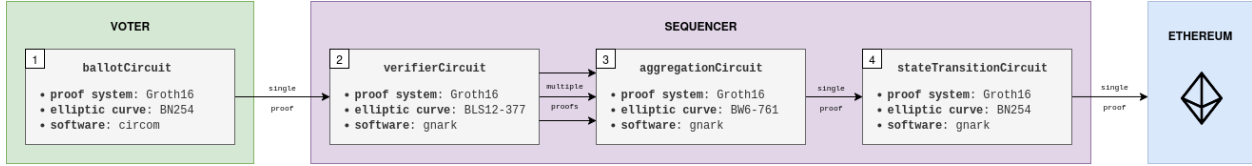


Fig. 3. Overview of the circuit flow in DAVINCI.

For clarity, the circuits are described in this section with their full set of public inputs explicitly listed. In the actual implementation, however, we apply an optimization: rather than exposing all public inputs individually, we compute a single commitment $h = \text{Hash}(PI)$ and use h as the only public input. The proof verifier then checks that $\text{Hash}(PI) = h$, ensuring that the original public inputs are consistent while reducing both proof size and verification cost. In DAVINCI, this commitment is computed using the MiMC-7 hash function (use [hlget](#) to point "Hash" to MiMC-7). Note that this optimization does not alter the semantics of the circuits but significantly reduces verifier complexity and on-chain verification costs. Other implementation aspects such as the exact number of constraints and the proving frameworks used are deferred to Section 8.2.

5.4.1 Ballot circuit

The ballot circuit, illustrated in Figure 4, is generated locally by the voter at the time of casting a ballot. Its purpose is twofold: first, to prove that the encrypted ballot is valid and complies with the protocol rules defined by the organizer; and second, to prove that the vote identifier (`voteID`) has been correctly derived. The correctness of this circuit is attested by a ZK proof generated using the ZK-SNARK protocol described in Section 4.7, instantiated over the BN254 curve. We call the resulting proof *ballot proof* and denote it as `ballotProof`. We detail below the inputs and constraints of the ballot circuit.

Public inputs

- **public** `processID`: voting process identifier.
- **public** `ballotMode`: ballot protocol configuration.S
- **public** `encryptionKey` (*EPK*): encryption public key.
- **public** `address`: voter's address.
- **public** `weight`: voter's weight.
- **public** (C_1, C_2) : encrypted ballot.
- **public** `voteID`: vote's unique identifier.

Private inputs

- **private** `ballot`: plaintext voter's voting choice.
- **private** k : random scalar.

Constraints

- *Vote encryption.* Correct encryption of the ballot: $(C_1, C_2) = \text{Enc}_{\text{encryptionKey}}(\text{ballot}; k)$.
- *Protocol rules compliance.* The ballot meets the protocol rules (number of valid choices, etc.): $\dots + \text{weight}$.
- *Vote identifier.* The vote's identifier is correctly computed: $\text{voteID} = \text{Hash}(\text{processID} || \text{address} || k)$.

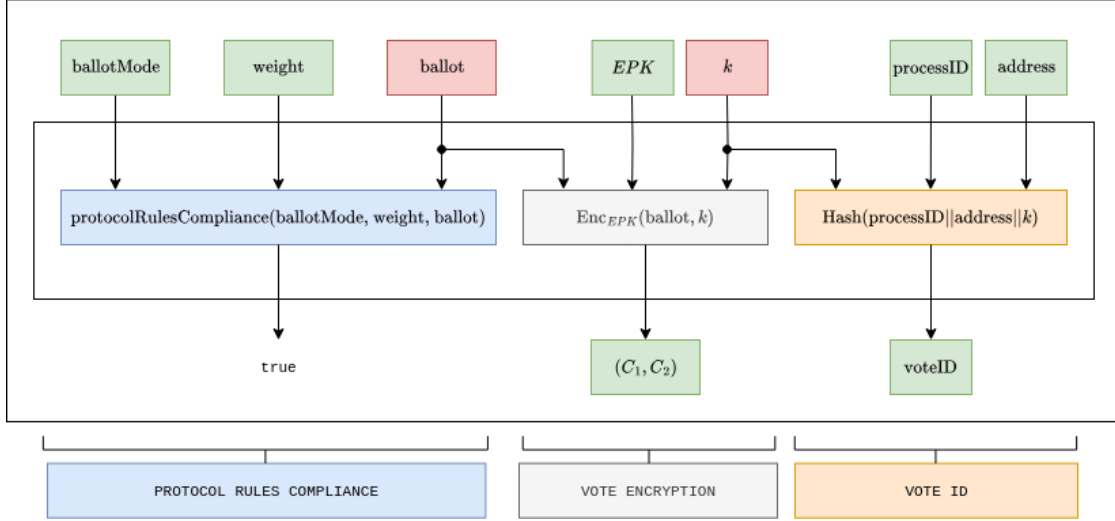


Fig. 4. Ballot circuit. All public values are framed in green.

5.4.2 Verifier circuit

The verifier circuit, illustrated in Figure 5, is generated by the sequencer when processing a vote. This circuit verifies the ballot's proof generated by the voter and it also validates the voter's eligibility in the census and the correctness of their digital signature. The correctness of this circuit is attested by a ZK proof generated using the ZK-SNARK protocol described in Section 4.7, instantiated over the BLS12-377 curve. We call the resulting proof the *authentication proof* and denote it as `authenticationProof`. We detail below the inputs and constraints of this circuit. – [Authentication of verification proof?](#)

Public inputs

- **public** processID: process identifier.
- **public** ballotMode: ballot protocol configuration.
- **public** encryptionKey (EPK): encryption public key.
- **public** (C₁, C₂): encrypted ballot.
- **public** voteID: voter's identifier.
- **public** censusRoot: census Merkle tree root.
- **public** pk: voter's public key.

Private inputs

- **private** ballotProof: voter's ballot proof.
- **private** weight: voter's weight.
- **private** censusProof: census membership proof.
- **private** signature: voter's signature.

Constraints

- *Voter's proof verification.* The vote zkProof is valid for the inputs provided:

$$P.Verify(ballotProof, PI = (processID, ballotMode, encryptionKey, (C_1, C_2), voteID, weight)).$$

- *Census membership.* The address derived from the user's public key is part of the census, and verifies the census proof with the user's weight provided:

$$MT^{census}.MembershipProof(weight, address = Hash_1(pk), censusProof, censusRoot).$$

- *Authentication + non-malleability.* The signature of the vote's identifier is valid for the given public key:

$$S.Verify_{pk}(voteID, signature).$$

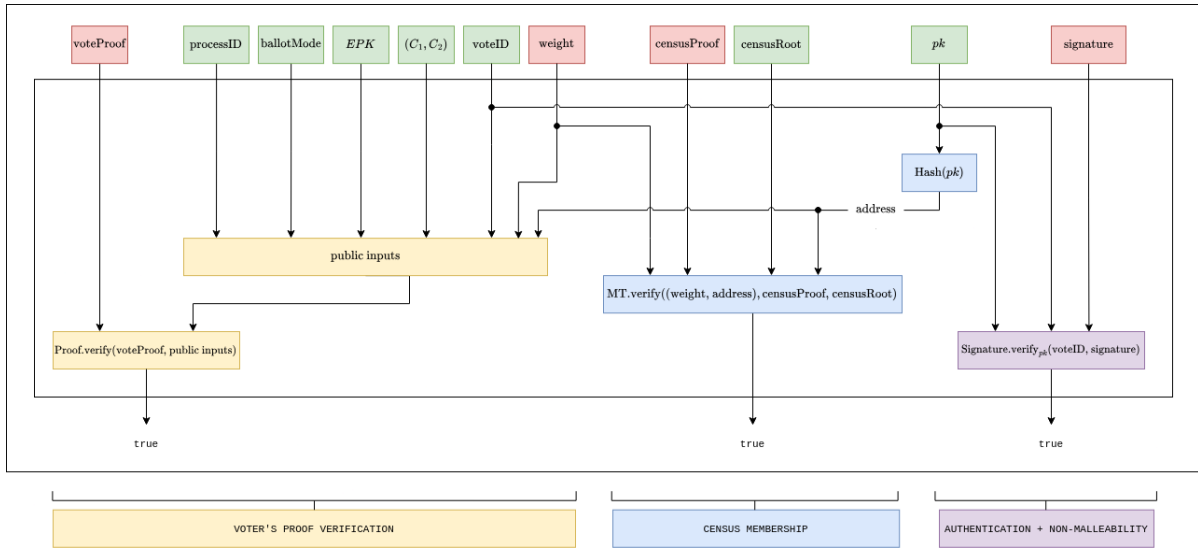


Fig. 5. Authentication circuit. All public values are framed in green.

5.4.3 Aggregation circuit

The aggregation circuit, illustrated in Figure 6, is generated by the sequencer to combine multiple authenticated votes into a single proof. Its purpose is to reduce verification overhead by recursively aggregating individual authentication proofs, while ensuring that all aggregated votes belong to the same voting process. The batch size, i.e., the maximum number of proofs aggregated in a single execution, is a fixed parameter (currently set to 60). If fewer proofs are available, the sequencer pads the batch with dummy proofs so that the circuit always operates on a fixed-size input. The correctness of this circuit is attested by a ZK proof generated using the ZK-SNARK protocol described in Section 4.7, instantiated over the BW6-761 curve. We call the resulting proof the *aggregation proof* and denote it as `aggregationProof`. We detail below the inputs and constraints of the aggregation circuit.

Public inputs

- **public** $(C_1, C_2)_i$: list of voters' encrypted ballots.
- **public** `voteIDi`: set of vote's identifiers.

- **public** pk_i : list of voters' public keys.
- **public** $processID$: process identifier.
- **public** $ballotMode$: ballot protocol configuration.
- **public** EPK : encryption public key.
- **public** $censusRoot$: census Merkle tree root.

Private inputs

- **private** $authenticationProof_i$: set of authentication proofs.

Constraints

- *Votes aggregation.* The accumulated authentication proofs are valid (for the given inputs). For every i :

$P.Verify(authenticationProof, PI = ((C_1, C_2)_i, voteID_i, pk_i, processID, ballotMode, EPK, censusRoot))$. ■

- *Shared public inputs.* The public inputs $processID$, $ballotMode$, $encryptionKey$, and $censusRoot$ are the same for all proofs.

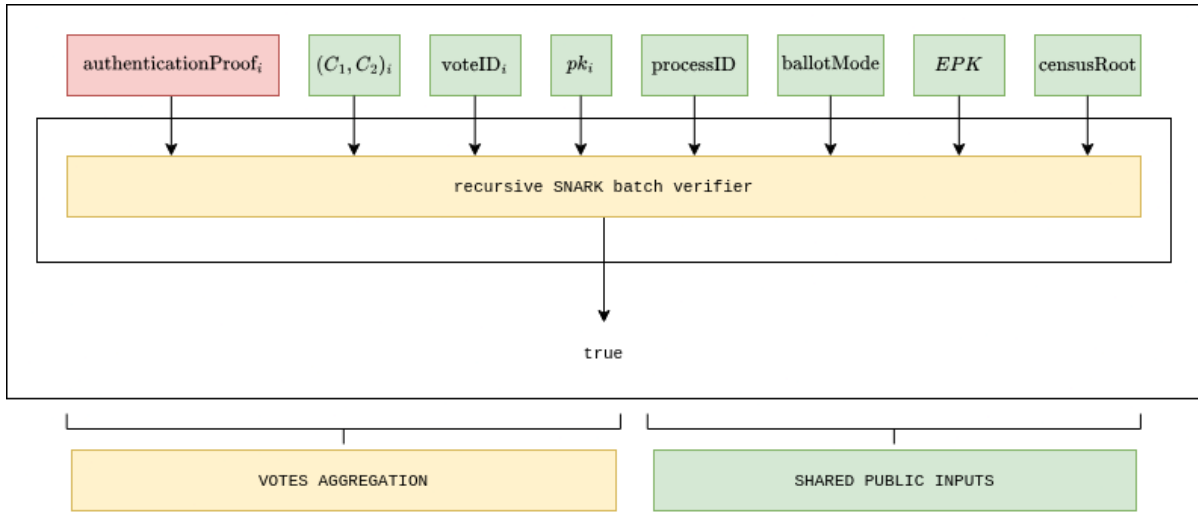


Fig. 6. Aggregation circuit. All public values are framed in green.

5.4.4 State transition circuit

The state transition circuit, illustrated in Figure 7, is generated by the sequencer to update the global state of the voting process. Its purpose is to verify that a batch of aggregated votes has been correctly incorporated into the state Merkle tree, that overwrites and vote identifiers are handled consistently, and that the accumulators of encrypted results are updated accordingly. The correctness of this circuit is attested by a ZK proof generated using the ZK-SNARK protocol described in Section 4.7, instantiated over the BN254 curve, which is supported by Ethereum's native precompiles for proof verification. We call the resulting proof the *state proof* and denote it as **stateProof**. This proof is submitted on-chain and verified by the process management smart contract, ensuring that every accepted state transition complies with the protocol rules.

Assertions.

- *Aggregation verification*: The aggregated zkProof is valid.
- *Transition verification*: The MerkleTree transition witness proves every change between Root1 and Root2.
- *Public inputs integrity*: ProcessID, BallotMode, CensusRoot, EncryptionKey remain unchanged.
- *Votes counting*: Ballots are correctly counted as new or overwrites, and added to results accumulators.

When a new voting process begins, the Sequencer initializes a new State, represented by the root hash of a Merkle tree. This tree encapsulates all essential information about the voting process, including process parameters, voter registry (census), ballot configurations, and initial results.

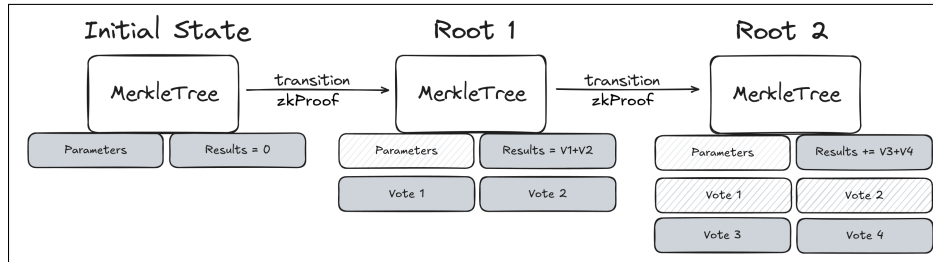
For each new batch of votes, the Sequencer updates the state by generating a zkSNARK proof that **validates the state transition from the current Root to a new Root**. This proof is submitted on-chain for settlement. By doing so, we maintain an immutable and verifiable record of the voting process on the blockchain.

This approach allows anyone to access the latest verified state of the voting process from the Results Verification smart contract, along with the necessary data to process subsequent state updates. The system's design enables multiple Sequencers to participate in tallying votes. They can take the current State Root and its associated data to construct the next state, incorporating new votes into the tally. This decentralization of Sequencers helps prevent potential censorship and reinforces the robustness of the voting process.

Maintaining the Chain. The chain of integrity is maintained through a combination of smart contract enforcement and strict zkSNARK circuit constraints. This ensures that each state transition is valid and builds upon the last accepted state without requiring additional mechanisms.

- **Sequential state roots**: Each state transition updates the Merkle tree from a previous root ('Root1') to a new root ('Root2') after processing a batch of votes.
- **Smart contract enforcement**: The smart contract verifies that the 'Root1' provided in the zkSNARK proof matches the last committed state root stored on-chain. This guarantees that all transitions are sequential and based on the latest accepted state.
- **Proof validation**: The smart contract uses the zkSNARK verification key to validate the submitted proof. A valid proof confirms that the transition from 'Root1' to 'Root2' adheres to all protocol rules enforced by the circuit.
- **State update**: Upon successful verification, the smart contract updates the stored state root hash to 'Root2', ensuring an immutable and continuous chain of states.

State Transition To validate and process state transitions, **we employ a zkSNARK circuit that enforces all protocol constraints**. This circuit proves that the transition from the previous state Root to the new state Root is valid based on the newly processed votes.



STATE TRANSITION CIRCUIT

Public inputs

- **public** *root_{old}*: Merkle tree root before the state transition.
- **public** *root_{new}*: Merkle tree root after the state transition.
- **public** *blobCommitment*: the commitment to the data blob containing the new votes.

Private inputs

- **private** Votes: The list of new votes being processed, including census proofs and authentication data.
- **private** Merkle Proofs of nullifier inclusion: Proofs that each voter nullifier is included in the census.
- **private** Merkle Proofs of results update: Proofs that the process results have been correctly updated.
- **private** Merkle Tree Update Witnesses: Necessary data (e.g., Merkle paths) to update the Merkle tree from Root1 to Root2.
- **private** Process Parameters: Retrieved from Root1 within the circuit.

Constraints

- **Immutable process parameters**: Ensure that critical process parameters (such as ‘censusRoot’ or ‘processId’) retrieved from ‘Root1’ remain consistent and are not altered in the transition.
- **Vote validity**: Validate that each vote proof is correct.
- **Voter eligibility**: Confirm that each voter is included in the census by verifying Merkle proofs of inclusion against the ‘censusRoot’ retrieved from ‘Root1’.
- **Nullifier non-existence**: Ensure that the nullifier for each vote does not exist in the current state (‘Root1’), preventing double voting.
- **Nullifier addition**: Correctly add each new nullifier to the state, resulting in ‘Root2’, updating the Merkle tree accordingly.
- **Results update**: Ensure that the voting results are accurately updated by adding the new votes to the previous results retrieved from ‘Root1’, so that ‘results2 = results1 + votes’.
- **Blob data integrity**: Confirm that the data used in the circuit (votes, nullifiers) corresponds to the ‘blobCommitment’ provided as a public input, ensuring that the votes processed are exactly those published in the data blob.
- **State transition validity**: Ensure that the new state root (‘Root2’) is correctly computed from ‘Root1’ by applying the validated votes and updates to the Merkle tree.

Marta: STATE TRANSITION CIRCUIT: a potential problem is that a sequencer may know if one of their votes has been overwritten, for this reason, sequencers reencrypt a certain amount of ballots every time (already existing ballots I mean). So, the circuit should include the reencryptions: both for existing ballots and also for the new votes, which should be reencrypted (for receipt-freeness).

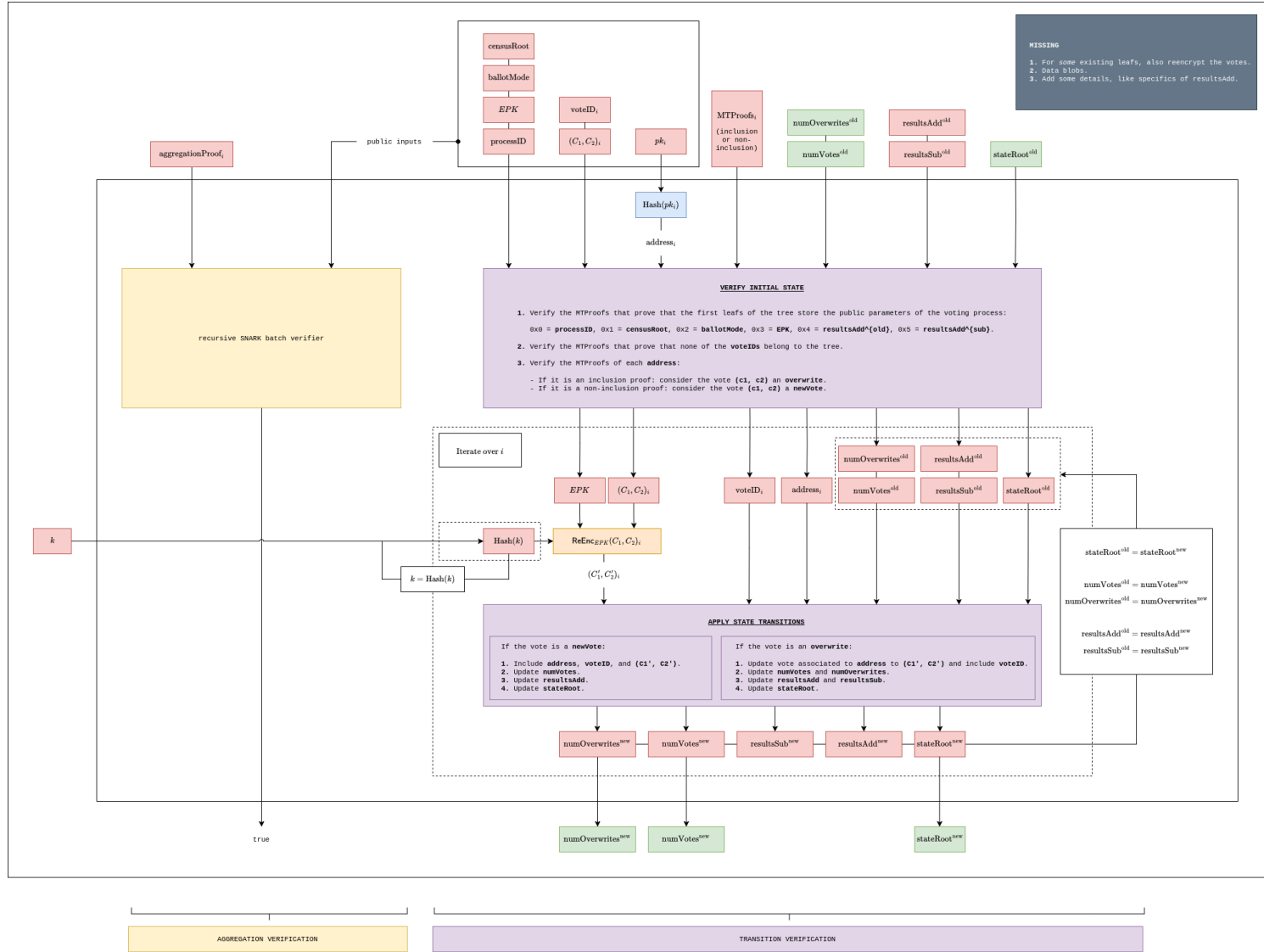


Fig. 7. State transition circuit. All public values are framed in green.

5.4.5 Results circuit

TBA.

5.5 Protocol flow

Marta: Rename steps: 1. voting setup, 2. encryption key generation, 3. votes collection, 4. results validation.

In Figure 8 we describe the protocol flow.

5.5.1 Prior to voting (only once)

sequencers – this process is always open:

1. Register this and that way.
2. Public keys? Sequencers must also publish how they can be reached (either when they register or when they contribute to the DKG). They can use a URI: Uniform Resource Identifier.
3. PROVIDE COLLATERAL. What happens to the collateral? If done only once, then you lose it? Do you need to send a new tx with more collateral?
4. Send transaction to Sequencer registry smart contract (pay for tx fee).

5.5.2 Start of the voting process

organizer

1. Prepares the census data `censusData` and generates census Merkle tree MT^{census} .
 2. Generates the census commitment: `censusRoot =  $MT^{\text{census}}$ .Root()`.
- Marta:** Additionally, after this step, the organizer should make a Merkle proof available to all eligible voters.
3. Set voting details: `duration, options, type, etc. + CENSUS MT ROOT`.
 4. Compute the process identifier `processID`, which is a unique 32-byte number derived from the organizer's Ethereum address and nonce, and the chain ID.
 5. Set security factors for DKG: a timeout for the DKG ceremony + minimum number of sequencers.
 6. Send transaction to the Ethereum process management smart contract with all data. Note this transaction has a cost (Eth tx fees) that needs to be covered by the organizer. The transaction includes the fields:

`processID, etc.`

5.5.3 Encryption key generation process

sequencers

1. Fetch the `processID`, `ballotMode`, and `censusRoot` from the process management smart contract.
2. Make contributions to DKG ceremony before the timeout set by the organizer.

After the timeout, all contributions to DKG are done. The smart contract checks if minimum number of sequencers is reached. If so, the `encryptionKey` “becomes available”.

Marta: The idea here is that some sequencer makes EPK available (a.k.a. sends a transaction). Alternatively, generate a SNARK proving the EPK has been generated correctly and the SC verifies it. It is not clear to me if it is sent to SR or PM.

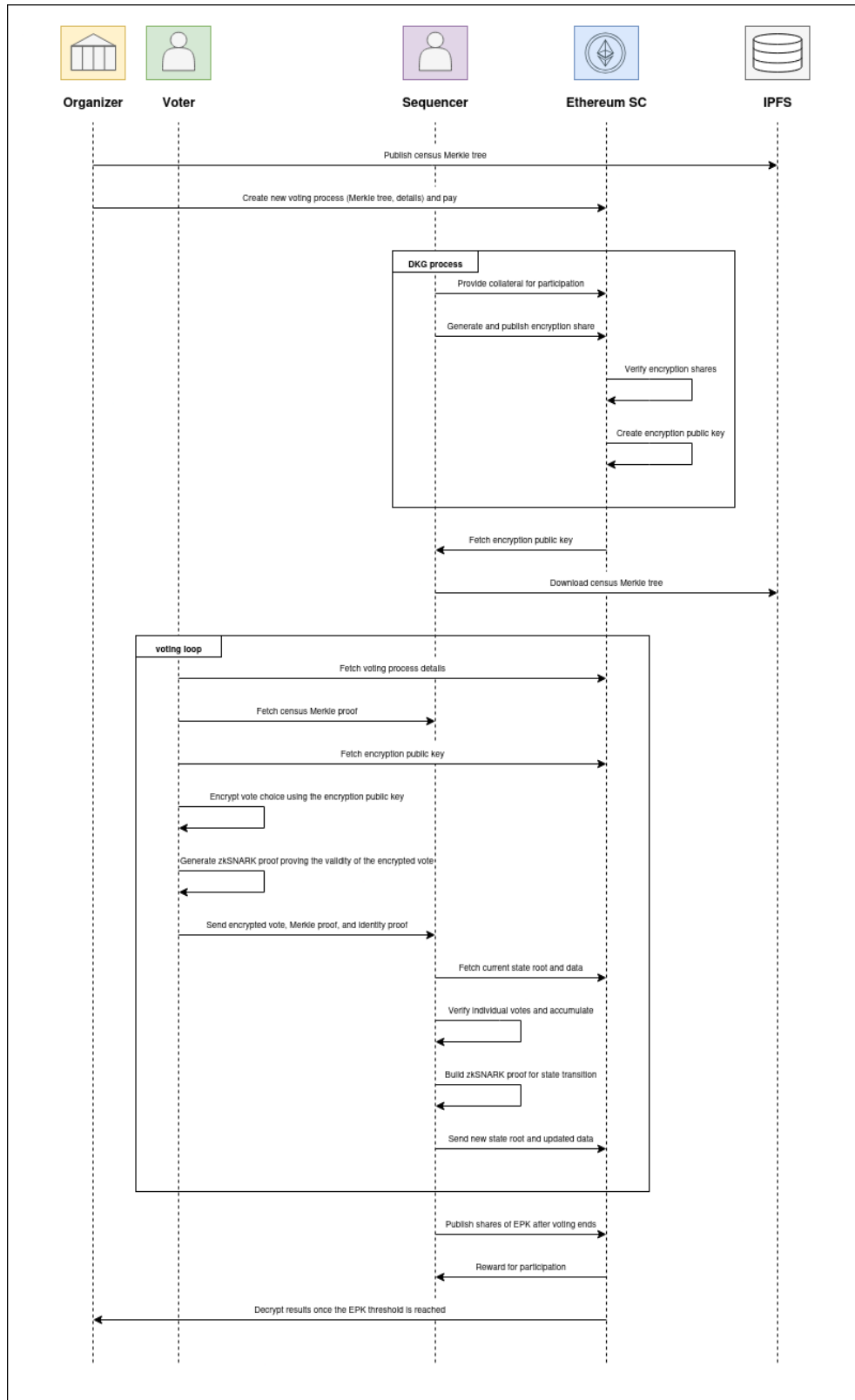


Fig. 8. Vocdoni voting process overview.

5.5.4 Voting process

voter

- Choose any of the available sequencers from the sequencer registry smart contract.
- Fetch from the census registry their membership proof `censusProof` to prove eligibility.
- Collect the `encryptionKey`, `ballotMode`, `processID` from the process management smart contract.
- Select their voting choice `ballot` according to the election rules specified by the `ballotMode` parameter.
- Sample a fresh random $k \in \mathbb{F}$ and encrypt the selected choice using the public key `encryptionKey`:

$$(C_1, C_2) = \text{Enc}_{\text{encryptionKey}}(\text{ballot}; k). \quad (1)$$

- Use the user's own Ethereum address `address` and the previous k to compute the `voteID`:

$$\text{voteID} = \text{MultiMiMC7}(\text{processID} \parallel \text{address} \parallel k). \quad (2)$$

- Generate ZK-SNARK of satisfiability of circuit from Fig. ?? with the proof system from Section 4.7 to prove that Eqs. (1) and (2) are correctly computed, and that `ballot` satisfies ballot protocol rules according to `ballotMode`.

$$\text{voteProof} = \text{Prove}(), \text{Circuit}(), \text{PublicInputs}.$$

- Sign `voteID` using the user's secret key `sk` to authenticates the vote and ensure that was cast by a legitimate voter:

$$\text{signature} = \text{S.Sign}_{\text{sk}}(\text{voteID}).$$

- Send the following data, which includes the encrypted vote, census membership proof, signature² and vote proof, to the selected sequencer:

$$\text{vote} = [\text{processID}, \text{voteID}, \text{encryptedBallot}, \text{censusProof}, \text{voteProof}, \text{signature}].$$

sequencers

- Fetch current state: retrieve the current valid process state root from the process management smart contract and the associated data from Ethereum blobs.
- $\text{censusRoot} = \text{MT}^{\text{census}}.\text{Root}()$.
- $\text{stateRoot} = \text{MT}^{\text{state}}.\text{Root}()$.
- Generate a SK-SNARK proof of state transition proving:
 - the validity of the ZK-SNARK proof provided by the voter,
 - the correct accumulation of votes from users, adding them to the process state,
 - the correct sum of the new encrypted votes using the homomorphic properties of ElGamal,
 - the new votes are from eligible users by checking the census Merkle proofs,
 - the new voters have not already voted by checking their nullifiers, or it is a correct vote overwrite,
 - the data blob hash matches the data used to verify the transition.
- Submit updated state: send the new state root to the smart contract and store the updated data in Ethereum blobs.

Sequencers keep accumulating votes and create state transition until the finalization of the process.

Marta: I understand this finalization is a deadline set by the organizer.

SC:

- Upon receiving a sequencer's transaction, the SC verifies the zk-SNARK proof provided by the sequencer.
- Ensure the origin root corresponds to the current stored state root.
- Confirm that the blob hash matches the one stored in Ethereum.

²Note that the address is not send because it can be recomputed from the signature.

5.5.5 Results validation

Once the voting is completed (I understand the timeout set by the organizer expires), the following things happen.

sequencers

- t out of n sequencers publish their shares corresponding to the EPK to the Ethereum smart contract.

Marta: specify which SC.

- When the threshold of shares is reached, the results can be decrypted by anyone.

Marta: mention that the SC computes the corresponding ESK, or that can be computed by anyone off-chain, since all info is public.

- Sequencers receive reward for their correct participants depending on the number of sequenced votes.

Marta: slash if they don't provide the right share? reward only dependent of sequenced votes? any relation to their participation in the computation of either EPK or ESK?

This way:

- After the voting period ends and the results are decrypted, anyone can verify the correctness of the final result.

Marta: explain how.

- Use the zkSNARK state proof and publicly available data on-chain to ensure the integrity and correctness of the entire voting process.

Marta: elaborate on this.

5.5.6 Finalization of the voting process

6 Ballot protocol

Marta: It is not clear what is the advantage of defining a protocol like this. Is it easy because it is easy to implement as an arithmetic circuit? Make the advantage clear.

Marta: Yes. The advantage is that we can define many types of voting systems (or recounting) with just these few variables. Moreover, we avoid if-else statements, which makes it very convenient to implement in-circuit. More info here: <https://docs.vocdoni.io/architecture/data-schemes/ballot-protocol.html#vocdoni-results-interpretation> (no multiple question supported).

Marta: It is called Ballot Mode because we define some modes (like quadratic, etc.) that can be chosen by the organizer. Essentially, each of this modes has a special configuration, but in order to make it easier, the organizer only needs to choose which of the modes (configs) to apply.

Marta: How is the voter's **weight** encoded as part of the ballot?

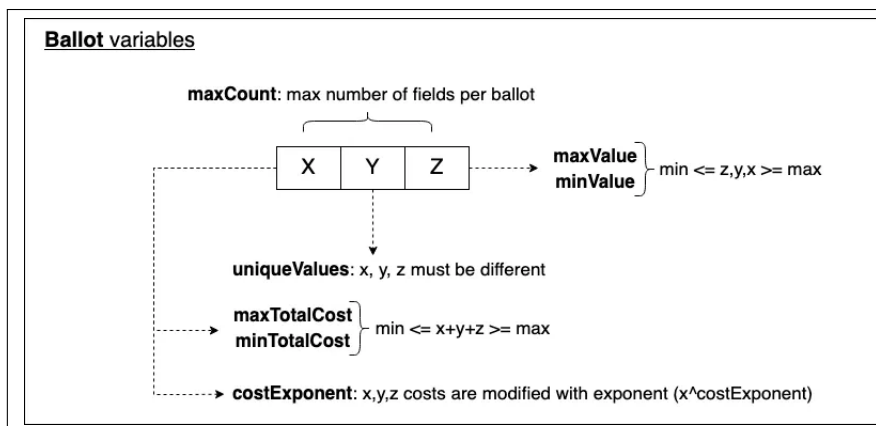
The DAVINCI ballot protocol is a simple and efficient mechanism for casting and tallying votes in any type of election or collective decision-making process. Each voting process consists of one or several fields and voters are required to provide a response for each of these fields in their ballot.

The responses in the ballot are represented as a sequence of natural numbers, each corresponding to the voter's choice for the respective field. Results are accumulated into a single array. Each position in the array corresponds to the sum of all votes cast for that field across all voters.

The ballot protocol is defined by a set of configurable variables that dictate how votes must be cast. This way the protocol can accommodate a wide range of voting processes and behaviors. The list of variables is the following.

1. **numFields**: it defines the maximum number of fields in a ballot (max 64).
2. **maxValue**: it is the maximum allowable value for any field in a ballot (if greater than 0).
3. **minValue**: it is the minimum allowable value for any field in a ballot (default 0).
4. **uniqueValues**: it specifies whether voters can select the same value multiple times within a ballot (default false).
5. **maxValueSum**: it limits the sum of all field values in a ballot (if greater than 0).
6. **minValueSum**: it specifies a minimum required total sum of field values in a ballot (default 0).
7. **costExponent**: it defines the exponent used to calculate the "cost" of votes for each field (default 1).

Marta: "if greater than 0" vs. "default" (make it consistent).



Marta: In Figure: when is says uniqueValues: x, y, z must be different – this is if uniqueValues = false, right?

Example 1: Rating candidates.

Marta: Reorder names Lennon, Hendrix, Joplin alphabetically.

Marta: In this example, the rest of variables are default, right? E.g. `minValue = 0`?

Consider a voting process where voters are asked to rate three candidates: Lennon, Hendrix, and Joplin. Voters rate each candidate from 0 to 5 stars, and each vote is represented as an array where each position corresponds to the candidate's rating. The configuration of this voting would be the following:

- `numFields`: 3
- `maxValue`: 5
- `uniqueValues`: Yes

Ballots:

- Vote 1: [3, 2, 5] (3 stars for Lennon, 2 stars for Hendrix, 5 stars for Joplin)
- Vote 2: [4, 3, 2]
- Vote 3: [2, 4, 5]

Marta: I would write 4 ballots.

After accumulating the votes:

- Results Array: $[3+4+2, 2+3+4, 5+2+5] = [9, 9, 12]$
Lennon received 9 points, Hendrix received 9 points, and Joplin received 12 points.

Marta: In the above example, there is no limit to `maxTotalCost`, right? Maybe add something like, if `maxTotalCost = X`, then one of the previous ballots would be invalid (and explain why).

Example 2: Quadratic voting for resource allocation. In a scenario where voters distribute a fixed number of credits across different options (e.g., selecting funding levels for NGOs), the ballot allows voters to assign multiple points, but the cost of casting multiple votes for a single option increases quadratically.

Configuration:

- `numFields`: 4
- `maxValueSum`: 12 (credits)
- `costExponent`: 2 (quadratic)

Ballots:

- Vote 1: [2, 2, 2, 0]
- Vote 2: [1, 1, 3, 1]
- Vote 3: [0, 2, 1, 2]

Marta: Add a fourth vote, e.g. [3, 3, 0, 0]. Then mention that the first 3 ballots are ok because $2^2 + 2^2 + 2^2 = 12$, $1^2 + 1^1 + 3^2 + 1^2 = 12$, $2^2 + 1 + 2^2 = 9 < 12$, but $3^2 + 3^2 = 18 > 12$, and therefore it's invalid. Did I get this right?

After accumulating the votes:

- Results Array: $[2+1+0, 2+1+2, 2+3+1, 0+1+2] = [3, 5, 6, 3]$
Each position in the array represents the total sum of credits allocated to each NGO.

Marta: Cost may be dependant on the voter's weight.

7 The Vocdoni token

DAVINCI introduces the Vocdoni token (VOC) as a key element of its decentralized voting ecosystem, playing a crucial role in the protocol's sustainability. The token serves multiple utility functions that align the incentives of all participants (voting organizers, sequencers, and voters) ensuring the integrity, efficiency, and security of the voting system.

7.1 Roles of the token

1. **Collateral for sequencers:** Sequencers are required to stake VOC tokens as a collateral to participate in the protocol. This serves as a safeguard to ensure responsible participation. If a sequencer behaves improperly (whether due to malicious intent or unintentional errors) it can face penalties, including the loss of part of its staked tokens.
2. **Incentive mechanism:** Sequencers earn rewards in VOC tokens based on their contribution to processing valid votes and maintaining the network. Rewards are proportional to the number of valid votes successfully added to the shared state.
3. **Payment for voting processes:** Voting processes organizers use VOC tokens to cover the costs of creating and managing voting processes. The costs depend on factors like the size of the voting registry, the voting period's duration, and the desired level of security (based on the number of participating sequencers).
4. **Governance:** The VOC token facilitates decentralized governance by giving the token holders the right to participate in the project governance. Token holders can influence on important matters such as protocol upgrades, ecosystem development, and other initiatives aimed to enhance various aspects of DAVINCI. This ensures that the project evolves in a transparent, community-driven manner.

7.2 Economics for organizers

Organizers of voting processes pay fees in VOC tokens to create and manage their voting events. These fees cover operational costs and incentivize the sequencers.

The costs of voting processes vary based on the following factors:

- **Maximum number of votes:** Larger voter registries require more resources for processing.
- **Voting duration:** Longer voting periods demand extended resource commitments from sequencers.
- **Security level:** Organizers can adjust the number of participating sequencers to balance between cost and security needs.

Fees must be paid upfront but can be partially reimbursed. The formula for calculating the reimbursement is:

$$\text{reimbursement} = \text{totalCost} - \text{totalReward} - \text{baseCost}$$

Given that there is a list of eligible voters for each voting process, organizers should reserve space equal to the maximum number of voters, anticipating that all eligible voters may participate. However, since this is unlikely, a portion of the reward pool may be reimbursed.

The components of this formula are defined in the following sections.

7.3 Voting process cost model

The process cost model defines how the cost for a given process is calculated. Considering sequencer-specific capacities, process duration, number of voters and security costs, we define following components formula:

$$\text{totalCost} = \text{baseCost} + \text{capacityCost} + \text{durationCost} + \text{securityCost}$$

7.3.1 Components definition

- **baseCost:** The base cost is a fixed fee charged by the sequencer to set up a voting process based on a fixed value plus the number of and a fixed factor. It is independent of the process duration, or the security level. This part cannot be reimbursed thus is a portion of the that will always be rewarded to the sequencers.

$$\text{baseCost} = \text{fixedCost} + \text{maxVotes} \cdot p$$

Where:

- **fixedCost** is a fixed base fee defined into the protocol.

- **maxVotes** is the maximum number of votes of a given voting process.
- p is a small lineal factor defined into the protocol.
- **capacityCost**: It defines the cost given the current occupancy of the sequencer network. This component models the cost of reserving space for voting processes, relative to the number of sequencers available, the number of running voting processes and the maximum numbers of voters. The cost increases non-linearly as the number available sequencers approaches to the total number of sequencers and the number of running voting processes grows, so when there are fewer sequencers available and big number of voting processes running, the capacity becomes more valuable.

$$k_1 \cdot \left(\frac{\text{totalVotingProcesses}}{\text{totalSequencers} - \text{usedSequencers} + \epsilon} \cdot \text{maxVotes} \right)^a$$

- k_1 : A scaling factor controlling the impact of sequencers usage.
- **totalVotingProcesses**: The total number of running voting processes.
- **totalSequencers**: The total number of registered sequencers.
- **usedSequencers**: The number of sequencers that are already handling other voting processes.
- a : An exponent that controls how sharply the cost increases as the used capacity approaches the total available capacity. In other words, it controls the non-linearity of the cost increase.
- ϵ : Is a very small number that avoids zero division.
- **durationCost**: This component models the cost of running the process for a specific duration, where longer processes incur into more cost. The scaling is non-linear, meaning that shorter processes are more cost-efficient, while longer processes are increasingly expensive. It is expected to have a minimum and maximum duration thresholds (e.g 1 hour to 1 year).

$$k_2 \cdot \text{processDuration}^b$$

- k_2 : A scaling factor for the process duration.
- **processDuration**: The duration of the process in hours.
- b : An exponent that controls the non-linear scaling of the process duration.
- **securityCost**: The security cost models the use of multiple sequencers to ensure a secure process. The cost scales exponentially based on the number of sequencers used, but with diminishing returns as the number of sequencers approaches the total available sequencers:

$$k_3 \cdot e^{c \cdot \left(\frac{\text{numSequencers}}{\text{totalSequencers}} \right)^d}$$

- k_3 : A scaling factor controlling the overall weight of security cost.
- c : Controls the steepness of the exponential scaling for security costs.
- **numSequencers**: The number of sequencers used in this process.
- **totalSequencers**: The total number of available sequencers in the network.
- d : An exponent controlling how fast the security cost increases as the number of sequencers approaches the total available sequencers.

7.3.2 Constraints

There is the need to enforce some constraints in the formula in order to avoid impracticable scenarios.

- If $\text{processDuration} > \text{maxDuration}$ then: $\text{totalCost} = \infty$.
- If $\text{numSequencers} > \text{totalSequencers}$ then: $\text{totalCost} = \infty$.

7.4 Economics for sequencers

To become a sequencer and earn rewards, participants must stake VOC tokens as collateral. This collateral is locked during the sequencer registry in the Sequencer Registry smart contract and can be withdrawn upon the sequencer commitments are fulfilled.

Sequencers **receive rewards** based on:

- The number of **votes** they include in the shared state.
- The number of **vote rewrites** they include in the shared state.
 - A vote rewrite can be either a vote overwrite (a voter casting another vote that overwrites the previous one) or a vote re-encryption (made by the sequencer). Rewriting votes enables more flexibility for the voter and is a key mechanism for the receipt-freeness.
 - It is expected for the sequencers to maximize the number of vote rewrites because there is no way to distinguish between a vote overwrite and a vote re-encryption. This is considered a positive behavior because it contributes to the receipt-freeness mechanism of the protocol.
 - A vote and a vote rewrite are distinguishable because the former include a nullifier that has not ever been included in a specific process.
 - Given that each sequencer will submit a ZK proof validating the state transition into the settlement layer smart contract it is straight forward to count the number of votes and the number of vote rewrites processed by each sequencer.
 - Sequencers can allow vote rewrites up to T times the number of new votes for any state transition. T will be defined as a constant in the protocol.
- The number of **non-processed votes** in relation to the maximum number of voters.

The total reward obtained can be expressed as:

$$\text{sequencerReward}_i = R \cdot \left(\frac{\text{votes}_i}{\text{maxVotes}} \right) + W \cdot \left(\frac{\text{voteRewrites}_i}{\text{totalRewrites}} \right)$$

And subject to the constraints:

$$\frac{\text{voteRewrites}_i}{\text{votes}_i} \leq T$$

$$\text{totalReward} = R + W$$

$$R > W$$

- $R > W$ since the sequencers must prioritize processing votes and not be incentivized to just rewrite existing votes.

Where:

- R : A part of the reward pool allocated for a specific voting process.
- votes_i : The number of votes processed by the sequencer for a specific voting process.
- maxVotes : The maximum number of voters participating in a voting process.
- W : A part of the reward pool allocated for a specific voting process.
- voteRewrites_i : The number of vote rewrites processed by the sequencer for a specific voting process.
- totalRewrites : The total number of vote rewrites for a specific process.

7.4.1 Penalties

- Non-participation: Sequencers who fail to meet their obligation, not providing key shares during the tally phase, can have their collateral slashed. Penalties for a sequencer can be expressed as:

$$\text{slashedAmount}_i = s \cdot \text{stakedCollateral}_i$$

Where:

- slashedAmount_i : The total slashed amount for a given sequencer.
- s : The slashing coefficient $0 \leq s \leq 1$.
- $\text{stakedCollateral}_i$: The amount of VOC tokens staked by a given sequencer.

7.5 Summary

The total cost formula combines four main components:

1. A base cost for setting up the process.
2. A capacity cost based on the number of voters, the total running processes and available sequencer capacity.
3. A duration cost based on how long the process lasts.
4. A security cost based on the number of sequencers used, with diminishing returns for using more than a certain number.

The proposed formula ensures that:

- Small processes are cost-efficient.
- Larger processes or processes using a high proportion of available sequencer capacity incur higher costs.
- The cost of security increases rapidly if more sequencers are used, but adding sequencers beyond a certain point leads to diminishing returns.
- Impractical scenarios cannot be reached.

7.5.1 Notes on optimization

In the model presented we can clearly see that there are conflicting objectives from the two main actors:

- The organizer (the “buyer” of services) wants to minimize the cost of running the process.
- The sequencers (the “sellers” of capacity) want to maximize their rewards. A sequencer can decide whether to participate based on expected profits.

These are naturally conflicting objectives that can be modeled as a strategic game. However, modeling this equilibrium for the protocol it is not in the scope on this document and will be presented in a separate piece.

8 Analysis

Marta: THE PROTOCOL (does not reveal the vote but it) REVEALS IF SOMEONE HAS VOTED OR NOT.

Marta: Check EthResearch post: <https://ethresear.ch/t/vocdoni-protocol-enabling-decentralized-voting-for-the-masses-with-zk-technology/21036>.

Marta: This section is old text, it needs to be reviewed.

8.1 Security discussion

Make a section called *properties*?

From old intro:

Design principles. To build the DAVINCI stack, we adhere to the following design principles:

1. **Cryptography as the source of truth:** we rely exclusively on cryptographic proofs to ensure the integrity and security of the voting process. By trusting only in cryptography, we eliminate the need for centralized authorities, making the system inherently secure and transparent.
2. **Trustlessness:** our system operates without requiring trust in any single party. Through cryptographic protocols and decentralized infrastructure, we ensure system integrity and prevent compromise from any malicious actor.
3. **End-to-end verifiability:** every voter can verify their ballot from casting to result computation (individual verifiability). Additionally, any third party can audit the election data to confirm results (universal verifiability) and verify that each vote comes from a uniquely registered voter (eligibility verifiability). Transparent cryptographic mechanisms make this possible.

4. **Composability:** the system is modular, consisting of interchangeable components that can be rearranged or integrated with external systems via adaptable interfaces. This allows for redundancy, flexibility, and seamless integration with third-party applications, exemplified by our voting-as-a-service APIs.
5. **Accessibility:** Vocdoni’s voting platform (App) is open source, universally available and user-friendly. The interface is intuitive for all users, including those less familiar with technology, and accommodates voters that use assistive technologies like screen readers.
6. **Open source:** by releasing our code openly, we invite anyone to audit and contribute, enhancing security and fostering community engagement. Transparency prevents security through obscurity and accelerates innovation.
7. **Resilience:** we design for robustness against hardware failures, network outages, and censorship. Infrastructure decentralization and distributed ownership enhance system availability and resistance to attacks.
8. **Scalability:** our solution processes votes at high throughput, exceeding current requirements to accommodate future growth. The decentralized infrastructure scales organically as usage increases, ensuring consistent performance.
9. **Receipt-freeness:** to mitigate risks of collusion, coercion, and vote-buying, the system enables voters to verify their votes without being able to prove to others how they voted, thus reducing incentives for coercion and bribery. Additionally, voters are allowed to overwrite their votes while maintaining secrecy.
10. **Automation:** we minimize human intervention through smart contracts and cryptographic protocols, reducing costs and human error. Automation ensures consistent operation and frees resources for voter support and auditing.

8.1.1 Receipt-freeness

In democratic processes, the ability of voters to cast their votes freely, without undue influence or coercion, is crucial. A critical aspect of maintaining this freedom is ensuring receipt-freeness, preventing voters from being able to prove to third parties how they voted. DAVINCI implements receipt-freeness by leveraging the properties of the ElGamal cryptosystem, and zkSNARKs.

Ballot re-encryption. The ElGamal cryptosystem over elliptic curves supports additive homomorphism, allowing operations to be performed on ciphertexts that translate to addition on the underlying plaintexts. Re-encryption is a process that refreshes the randomness of a ciphertext without changing the underlying plaintext message. This makes computationally infeasible to link the original and re-encrypted ciphertexts.

Handling receipts. To prevent voters from being able to prove how they voted, DAVINCI employs re-encryption of ballots by the Sequencers. When a voter submits an encrypted ballot, the Sequencer re-encrypts it before storing it in the state Merkle tree.

This way, the system ensures that voters cannot produce a receipt of their vote by revealing r since the stored ciphertext no longer corresponds to r . This prevents vote-buying and coercion by third parties.

Handling collusion. To further enhance receipt-freeness and mitigate the risk of collusion between voters and Sequencers, DAVINCI allows voters to overwrite their votes. A voter can submit multiple votes, with each new submission replacing the previous one.

When a Sequencer receives a new vote from a voter who has already voted, it performs the following steps:

1. **Detect overwrite:** The Sequencer checks the state Merkle tree using the voter’s nullifier to determine if the voter has previously cast a vote.
2. **Subtract previous vote:** The Sequencer takes the existing encrypted ballot and adds it to a "Subtractive Results" accumulator. This effectively negates the previous vote in the final tally.
3. **Add new vote:** The new encrypted ballot is added to the "Results" accumulator.
4. **Update state Merkle tree:** The Sequencer re-encrypts the new ballot and updates the state Merkle tree with this re-encrypted ballot.

Concealing vote overwrites. To prevent observers from detecting when overwrites occur, the Sequencers regularly re-encrypt a random subset of ballots in the state Merkle tree during each state update. This process obscures the occurrence of overwrites, as re-encryptions are indistinguishable from standard re-randomizations performed for privacy enhancement.

By re-encrypting random ballots, the system increases the entropy and makes it statistically improbable for an adversary to determine if a specific ballot was overwritten or simply re-randomized.

8.1.2 Privacy

DAVINCI ensures user anonymity by anonymizing both the ballot and the voter’s identity, employing cryptographic techniques that maintain privacy even in the face of future quantum computing threats. The system uses the ElGamal homomorphic encryption scheme to encrypt ballots, allowing for the aggregation of votes without revealing individual choices. Since encrypted ballots are stored in public repositories like Ethereum blobs—which, although removed after some weeks, may still be accessible—it is crucial to prevent any association between decrypted ballots and voter identities. By decoupling the voter’s identity from their encrypted ballot through the use of secrets and cryptographic hashes, even if an adversary decrypts the ballots in the future, they cannot link them back to individual voters.

The identity anonymization acts as a double security factor, enhancing long-term privacy. While the Sequencer processing the vote can identify the voter (since voters submit proofs of eligibility and commitments), there is no incentive for the Sequencer to make this information public, and the Sequencer cannot decrypt the voter’s ballot because they do not possess the private decryption keys. This ensures that voter choices remain confidential.

We have adopted this partial identity anonymization because generating fully anonymous proofs using zkSNARKs directly from digital signatures (e.g., ECDSA/EdDSA or RSA) is computationally intensive for client-side devices like browsers and smartphones. Our priority is to support a wide range of devices, making the system accessible to as many voters as possible. In the future, as cryptographic technology advances and client devices become more powerful, we anticipate being able to generate such zero-knowledge proofs efficiently on the client side. This would enable us to fully anonymize the client’s identity in addition to the ballot, further enhancing user privacy without compromising accessibility or user experience.

8.1.3 Quantum resistance

As quantum computing technology advances, it poses significant challenges to classical cryptographic schemes that underpin the security of digital systems, including voting platforms like DAVINCI. Ensuring that the system remains secure in the face of quantum threats is crucial for its longevity and trustworthiness.

Quantum computers have the potential to solve certain mathematical problems exponentially faster than classical computers. Notably, Shor’s algorithm allows quantum computers to efficiently factor large integers and compute discrete logarithms, undermining the security of widely used cryptographic schemes such as RSA, DSA, ECDSA, and the ElGamal cryptosystem.

However, the design of DAVINCI incorporates mechanisms that preserve voter anonymity even in the face of future quantum attacks:

- Detachment of identity and encrypted Ballot: The voter’s identity and their encrypted ballot are decoupled through the use of a secret in the nullifier. The nullifier is computed as:

$$N = \text{Hash}(\text{processID} || s)$$

where s is a secret known only to the voter. This means that even if an adversary decrypts the encrypted ballots using a quantum computer, they cannot link a decrypted vote back to a voter’s identity in the census without knowledge of the secret s .

Marta: Instead of “detachment”, I propose “unlinkability”: identity unlinkability.

- Quantum-resistant hash functions: the `nullifier` and `identityCommitment` are computed using cryptographic hash functions that are believed to be resistant to quantum attacks (e.g., SHA-3). While Grover’s algorithm can provide a quadratic speedup in searching for preimages, using sufficiently long hash outputs (e.g., 256 bits) mitigates this risk.

To further safeguard against quantum threats, the following measures can be implemented in the future:

1. Adopt post-quantum signature schemes. Replace ECDSA/EdDSA with quantum-resistant algorithms such as CRYSTALS-Dilithium, Falcon, or Rainbow, which are based on hard lattice problems.
2. Explore lattice-based homomorphic encryption schemes. Replace ElGamal cryptosystem with quantum-resistant alternatives such as the Brakerski-Gentry-Vaikuntanathan (BGV) or the Brakerski/Fan-Vercauteren (BFV) schemes.
3. Adopt quantum-resistant zero-knowledge proof systems, such as zkSNARK constructions based on post-quantum assumptions or **zkSTARKs**.

Marta: Citations to protocols are missing.

Marta: If it is not quantum resistant, I would leave it to future work. Although unconventional, we could also make a “protocol weaknesses” section.

8.1.4 Data availability

8.2 Implementation

Marta: Link to repos, etc. + details such as circuits have been implemented using `circom`, this and that.

Circuits.

- Circuit 1: `circom/snarkJS`, ~ 53.000 constraints.
- Circuit 2: `gnark`, ~ 3.1 million constraints.
- Circuit 3: `gnark`, $40.000 \times (\text{number of votes})$ constraints.
- Circuit 4: `gnark`, ~ 16 million constraints.

8.3 Performance evaluation

Marta: E.g. circuit constraints benchmarks.

9 Conclusions

10 Future work

- Voter: they do `circuit1` + `circuit2` themselves (instead of the sequencer doing `circuit2`).
- Post-quantum.
- Support to xxx.

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References

1. Albrecht, M., Grassi, L., Rechberger, C., Roy, A., Tiessen, T.: Mimc: Efficient encryption and cryptographic hashing with minimal multiplicative complexity. In: Cheon, J.H., Takagi, T. (eds.) *Advances in Cryptology – ASIACRYPT 2016*. pp. 191–219. Springer Berlin Heidelberg, Berlin, Heidelberg (2016)
2. Baylina, J., Bellés, M.: Sparse Merkle trees (2019), <https://docs.iden3.io/publications/pdfs/Merkle-Tree.pdf>
3. Bellés-Muñoz, M., Whitehat, B., Baylina, J., Daza, V., Muñoz-Tapia, J.L.: Twisted edwards elliptic curves for zero-knowledge circuits. *Mathematics* **9**(23) (2021). <https://doi.org/10.3390/math9233022>, <https://www.mdpi.com/2227-7390/9/23/3022>
4. Bertoni, G., Daemen, J., Peeters, M., Van Assche, G.: The KECCAK SHA-3 submission (2011), <https://keccak.team/files/Keccak-submission-3.pdf>
5. Blake, I.F., Seroussi, G., Smart, N.P.: *Elliptic curves in cryptography*. Cambridge University Press, USA (1999)
6. Bowe, S., Chiesa, A., Green, M., Miers, I., Mishra, P., Wu, H.: ZEXE: enabling decentralized private computation. In: 2020 IEEE Symposium on Security and Privacy, SP 2020, San Francisco, CA, USA, May 18-21, 2020. pp. 947–964. IEEE (2020). <https://doi.org/10.1109/SP40000.2020.00050>, <https://doi.org/10.1109/SP40000.2020.00050>
7. Brown, D.R.L.: SEC 2: Recommended elliptic curve domain parameters. In: *Standards for efficient cryptography 2 (SEC 2)* (2010), <https://www.secg.org/sec2-v2.pdf>
8. El Housni, Y., Guillevic, A.: Optimized and secure pairing-friendly elliptic curves suitable for one layer proof composition. In: Krenn, S., Shulman, H., Vaudenay, S. (eds.) *Cryptology and Network Security*. pp. 259–279. Springer International Publishing, Cham (2020)
9. Grassi, L., Khovratovich, D., Rechberger, C., Roy, A., Schofnegger, M.: Poseidon: A new hash function for zero-knowledge proof systems. In: 30th USENIX Security Symposium (USENIX Security 21). pp. 519–535 (2021)
10. Groth, J.: On the size of pairing-based non-interactive arguments. In: *Annual international conference on the theory and applications of cryptographic techniques*. pp. 305–326. Springer (2016)
11. Housni, Y.E., Connor, M., Guillevic, A.: EIP-3026: BW6-761 curve operations [DRAFT], Ethereum Improvement Proposals, no. 3026, [online serial] (October 2020), <https://eips.ethereum.org/EIPS/eip-3026>
12. Jancar, J.: Standard curve database: BN254 (2020), <https://neuromancer.sk/std/bn/bn254>
13. Johnson, D., Menezes, A., Vanstone, S.: The elliptic curve digital signature algorithm (ECDSA). *Int. J. Inf. Secur.* **1**(1), 36–63 (August 2001). <https://doi.org/10.1007/s102070100002>, <https://doi.org/10.1007/s102070100002>
14. Sutikno, S., Surya, A., Effendi, R.: An implementation of elgamal elliptic curves cryptosystems. In: IEEE. APC-CAS 1998. 1998 IEEE Asia-Pacific Conference on Circuits and Systems. Microelectronics and Integrating Systems. Proceedings (Cat. No.98EX242). pp. 483–486 (1998). <https://doi.org/10.1109/APCCAS.1998.743829>
15. Vlasov, A., hujw77: EIP-2539: BLS12-377 curve operations [DRAFT], Ethereum Improvement Proposals, no. 2539, [online serial] (February 2020), <https://eips.ethereum.org/EIPS/eip-2539>
16. WhiteHat, B., Bellés, M., Baylina, J.: ERC-2494: Baby Jubjub elliptic curve [DRAFT], Ethereum Improvement Proposals, no. 2494, [online serial] (January 2020), <https://eips.ethereum.org/EIPS/eip-2494>
17. Wood, G., et al.: Ethereum: A secure decentralised generalised transaction ledger. *Ethereum project yellow paper* **151**(2014), 1–32 (2014)